Reading Notes on Framed Cobordisms

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Preface

These are my reading notes and final report for the Spring 2026 reading course on differential topology and cobordism theory. The course focused on *Topology from the Differentiable Viewpoint* by John Milnor and was supplemented by Daniel S. Freed's lecture notes *Bordism: Old and New*.

1 Orientation of Manifolds

We begin with the linear algebra underlying orientations.

Orientation of Vector Spaces. Orientations on a manifold are described by compatible orientations of its tangent spaces. We therefore begin with vector spaces. Let V be an n-dimensional real vector space. Define the set of ordered k-frames in V by

$$\mathrm{Fr}_k(V) := \{(v_1, \dots, v_k) \in V^{\times k} \mid v_1, \cdots, v_k \text{ linearly independent}\}$$

In particular, an n-frame in $\mathrm{Fr}_n(V)$ is an ordered basis of V and each n-frame corresponds to an element in the exterior algebra $\Lambda^n V$. Note that there is a natural $\mathrm{GL}^+(n,\mathbb{R})$ action on $\mathrm{Fr}_n(V)$ by change-of-basis matrices $[\mathrm{id}_V]_{\beta_1}^{\beta_2}$ for $\beta_1, \beta_2 \in \mathrm{Fr}_n(V)$.

Definition 1.1 (Orientation of Vector Spaces)**.** The set of orientations of an n-dimensional vector space V is the quotient of its frame space by the $\mathrm{GL}^+(n,\mathbb{R})$ action:

$$\mathrm{Or}(V) := \mathrm{Fr}_n(V)/\mathrm{GL}^+(n,\mathbb{R})$$

An orientation on V is an element $o \in \mathrm{Or}(V)$, and a frame is called positively oriented with respect to o if its $\mathrm{GL}^+(n,\mathbb{R})$ orbit is o.

The following proposition describes the equivalence relation in terms of the top exterior power.

Proposition 1.2. *Let $f_1, f_2 \in \Lambda^n V$ be the elements corresponding to n-frames $\beta_1, \beta_2 \in \mathrm{Fr}_n(V)$, $\dim V = n$. Then $\beta_1 \sim \beta_2$ if and only if $\exists c > 0 : f_1 = c f_2$.*

Proof. Recall that the top exterior power is given by

$$\Lambda^n V := V^{\otimes n}/I, \qquad I := \langle v \otimes v \in V^{\otimes 2} \mid \forall v \in V \rangle$$

For an n-frame, there is a natural map $\mathrm{Fr}_n(V) \hookrightarrow V^{\otimes n} \to \Lambda^n V \setminus \{0\}$ which sends

$$(v_1, \dots, v_n) \mapsto v_1 \wedge \cdots \wedge v_n \neq 0$$

The image is nonzero by the linear independence of a frame.
($\Rightarrow$) For n-frames $\beta_1 = (v_1, \dots, v_n)$ and $\beta_2 = (w_1, \dots, w_n)$ such that $\beta_1 \sim \beta_2$, let

$$f_1 = v_1 \wedge \cdots \wedge v_n, \qquad f_2 = w_1 \wedge \cdots \wedge w_n$$

Since $\dim V = n$, the top exterior power has dimension $\binom{n}{n} = 1$. Consider the unique linear isomorphism $A \in \mathrm{GL}^+(n,\mathbb{R})$ such that $A(w_i) = v_i$. This map induces an isomorphism on the top exterior power:

$$(\Lambda^n A)(f_2) = A(w_1) \wedge \cdots \wedge A(w_n) = v_1 \wedge \cdots \wedge v_n = f_1$$

By the multilinearity of the wedge product, this gives

$$f_1 = \det(A) \cdot f_2$$

Since $A \in \mathrm{GL}^+(n,\mathbb{R})$, we have $\det A > 0$, which proves the claim.
($\Leftarrow$) The converse follows from the same calculation: for any two bases β_1, β_2 of V in $\mathrm{Fr}_n(V)$, one can find the unique linear isomorphism A that sends β_1 to β_2. The calculation above shows that $\det A > 0$, so $A \in \mathrm{GL}^+(n,\mathbb{R})$, and hence f_1 and f_2 are in the same orbit. □

Thus, an orientation can also be described by a positive ray in the top exterior power, $\mathbb{R}_{>0} \cdot f \subset \Lambda^n V \setminus \{0\}$. This description is encoded by the determinant. As shown in the proof of Proposition 1.2, the determinant

$$\det : \mathrm{GL}(n,\mathbb{R}) \to \mathbb{R}^\times$$

determines the action $\mathrm{GL}(n,\mathbb{R}) \times \Lambda^n V \to \Lambda^n V$ given by $A \cdot f = \det(A) f$. In particular, the exterior algebra satisfies the following property.

Proposition 1.3. *The short exact sequence of finite-dimensional real vector spaces*

$$0 \longrightarrow V_1 \xrightarrow{\ i\ } V \underset{\pi}{\overset{s}{\rightleftarrows}} V_2 \longrightarrow 0$$

admits a splitting $s : V_2 \to V$. A choice of splitting induces an isomorphism of graded algebras

$$\Lambda^* V_1 \otimes \Lambda^* V_2 \xrightarrow{\cong} \Lambda V$$

where the left-hand side is the graded tensor product, the degree of which satisfies $|a \otimes b| = |a| + |b|$.

In other words, let V_1 and V_2 have dimensions n_1 and n_2, respectively, and write $V := V_1 \oplus V_2$. Then the exterior algebra satisfies

$$\Lambda^k V \cong \bigoplus_{a+b=k} \Lambda^a V_1 \otimes \Lambda^b V_2$$

In particular, the isomorphism in top degree is canonical: $\Lambda^{n_1+n_2} V \cong \Lambda^{n_1} V_1 \otimes \Lambda^{n_2} V_2$.

Proof. We construct the isomorphism from the short exact sequence. Choose a splitting $s : V_2 \to V$ with $\pi \circ s = \mathrm{id}_{V_2}$. For any $v \in V$, we have

$$\pi\big(v - (s \circ \pi)(v)\big) = \pi(v) - (\pi \circ s \circ \pi)(v) = 0$$

Hence $v - (s \circ \pi)(v) \in \ker(\pi) = i(V_1)$. Thus, there exists $v_1 \in V_1$ such that

$$v = i(v_1) + (s \circ \pi)(v)$$

On the other hand, if there is a nonzero vector $w \in i(V_1) \cap s(V_2)$, then by exactness, $\pi(w) = (\pi \circ i)(w_1) = 0$ for some $w_1 \in V_1$. Since $\pi \circ s = \mathrm{id}_{V_2}$, $s : V_2 \to V$ is injective and thus $w = s(0) = 0$. This shows that $i(V_1) \cap s(V_2) = \{0\}$, which means

$$V = i(V_1) \oplus s(V_2)$$

Let $A := \Lambda^* V_1 \otimes \Lambda^* V_2$. The inclusion $i : V_1 \hookrightarrow V$ and the splitting $s : V_2 \hookrightarrow V$ induce the map

$$\Phi_s : A \to \Lambda^*(V_1 \oplus V_2), \quad \Phi_s(\alpha \otimes \beta) = \Lambda^* i(\alpha) \wedge \Lambda^* s(\beta)$$

It remains to prove that Φ_s is a graded algebra isomorphism. Since $\Lambda^* i$ and $\Lambda^* s$ are induced homomorphisms, it suffices to construct an inverse to Φ_s. Define the linear map

$$j_s : V \to V_1 \oplus V_2 \cong A^1, \quad j(v) = \mathrm{pr}_1(v) \otimes 1 + 1 \otimes \mathrm{pr}_2(v)$$

Here $\mathrm{pr}_i : V \to V_i$ is the projection associated with the chosen direct sum decomposition, and the degree-one part $A^1 = (\Lambda^1 V_1 \otimes \mathbb{R}) \oplus (\mathbb{R} \otimes \Lambda^1 V_2) \cong V_1 \oplus V_2$. Since j_s takes values in the degree-one part of A, it satisfies $j_s(v)^2 = 0$. By the universal property of the exterior algebra, it extends uniquely to a graded algebra homomorphism

$$\Psi_s : \Lambda^* V \to A, \quad \Psi_s|_{\Lambda^1 V} = j_s$$

Explicitly, the map is defined by $\Psi_s(v) = \mathrm{pr}_1(v) \otimes 1 + 1 \otimes \mathrm{pr}_2(v)$. We check that $\Psi_s = \Phi_s^{-1}$:

1. For any $v_1 \in V_1$, $\Psi_s\big(i(v_1)\big) = \mathrm{pr}_1\big(i(v_1)\big) \otimes 1 + 1 \otimes \mathrm{pr}_2\big(i(v_1)\big) = v_1 \otimes 1$.
2. For any $v_2 \in V_2$, $\Psi_s\big(s(v_2)\big) = \mathrm{pr}_1\big(s(v_2)\big) \otimes 1 + 1 \otimes \mathrm{pr}_2\big(s(v_2)\big) = 1 \otimes v_2$.

Thus $\Psi_s \circ \Phi_s$ is the identity on $V_1 \cong \Lambda^1 V_1 \otimes \mathbb{R}$ and $V_2 \cong \mathbb{R} \otimes \Lambda^1 V_2$. These subspaces generate A, so

$$\Psi_s \circ \Phi_s = \mathrm{id}_A$$

Conversely, for any $v \in V$, we have

$$\begin{aligned} \Phi_s \circ \Psi_s(v) &= \Phi_s\big(\mathrm{pr}_1(v) \otimes 1 + 1 \otimes \mathrm{pr}_2(v)\big) \\ &= i\big(\mathrm{pr}_1(v)\big) \wedge 1 + 1 \wedge s\big(\mathrm{pr}_2(v)\big) = v \end{aligned}$$

Since $\Lambda^* V$ is generated by the elements in V, $\Phi_s \circ \Psi_s = \mathrm{id}_{\Lambda^* V}$. Thus, $\Psi_s = \Phi_s^{-1}$, which completes the proof of the proposition. □

In particular, in top degree, let $n_i := \dim V_i$ for $i = 1, 2$ and $n = n_1 + n_2$. The short exact sequence naturally gives the canonical isomorphism

$$\Delta : \Lambda^{n_1} V_1 \otimes \Lambda^{n_2} V_2 \xrightarrow{\cong} \Lambda^n V, \quad \Delta(\xi \otimes \eta) = \Lambda^{n_1} i(\xi) \wedge \Lambda^{n_2} s(\eta)$$

The difference between the splittings $s, s' : V_2 \to V_2$ takes values in $\ker(\pi) = \mathrm{im}(i)$. Thus there is a linear map $T : V_2 \to V_1$ such that $s' = s + i \circ T$ and the top-degree isomorphism is given by

$$\Delta(\xi \otimes \eta) = \Lambda^{n_1} i(\xi) \wedge \Lambda^{n_2} s'(\eta) = \Lambda^{n_1} i(\xi) \wedge \Lambda^{n_2} s(\eta) + \Lambda^{n_1} i(\xi) \wedge \Lambda^{n_2} (i \circ A)(\eta)$$

where, by the definition of wedge product, the second term is zero. Proposition 1.3 gives the following corollary.

Corollary 1.4. *Given orientations $o_i \in \mathrm{Or}(V_i)$, the frame $(e_1, e_2) \in \mathrm{Fr}_n(V_1 \oplus V_2)$ is positively oriented if and only if e_i is positively oriented in V_i.*

Also, in $V_1 \oplus V_2$, exchanging the order of the basis will lead to a Koszul sign

$$(e_2, e_1) = (-1)^{n_1 n_2} (e_1, e_2)$$

which follows from the natural grading on $\Lambda^n V$.

We will also use the induced orientation on the dual space.

Proposition 1.5. *An orientation on a vector space V induces an orientation on its dual V^*.*

In the finite-dimensional case, this is best understood through the canonical determinant-line isomorphism $\Lambda^n(V^*) \cong (\Lambda^n V)^*$. An orientation of V^* is a choice of positive ray on $\Lambda^n(V^*)$, and the dual positive ray $(\Lambda^n V)^*$ corresponds canonically to the positive ray in $\Lambda^n V$ since $\dim \Lambda^n V = 1$. This definition is independent of the choice of a positively oriented basis. In the discussion of smooth manifold orientations, we often use cotangent spaces rather than tangent spaces.

Orientation of Manifolds. For a smooth manifold, recall that the tangent space at a point $p \in M$ is defined as

$$T_p M := \{ X : \mathcal{O}_{M,p} \to \mathbb{R} \mid X \text{ is a point derivation at } p \}$$

where $\mathcal{O}_{M,p} := C^\infty_{M,p}$ is the $\mathbb{R}$-algebra of smooth function germs at $p \in M$. An orientation on a smooth manifold gives compatible orientations on its tangent spaces. We first express this compatibility in terms of an atlas.

Definition 1.6 (Oriented Atlas)**.** Let M be a smooth n-manifold. A smooth atlas $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in A}$ is oriented if for any $\alpha, \beta \in A$ the transition map

$$\varphi_{\alpha\beta} := \varphi_\alpha \circ \varphi_\beta^{-1} : \varphi_\beta(U_\alpha \cap U_\beta) \to \varphi_\alpha(U_\alpha \cap U_\beta)$$

is an orientation-preserving diffeomorphism, i.e., its Jacobian determinant satisfies $\det D(\varphi_{\alpha\beta}) > 0$. A smooth manifold is orientable if and only if it admits an oriented atlas.

An oriented atlas determines an orientation on each tangent space. Indeed, if (U, φ) is a chart with local coordinates given by $\varphi = (x^1, \ldots, x^n)$, then for every $p \in U$, the tangent space $T_p M$ admits an ordered coordinate basis

$$\left(\frac{\partial}{\partial x^1}\Big|_p, \cdots, \frac{\partial}{\partial x^n}\Big|_p \right)$$

and we may declare this ordered basis to be positively oriented. If $p \in U_\alpha \cap U_\beta$, then the two coordinate bases are related by the Jacobian matrix of the transition map. Since the transition map has a positive Jacobian determinant, the two coordinate bases define the same orientation of $T_p M$. Hence, an oriented atlas gives a well-defined orientation on every tangent space $T_p M$. This leads to the following theorem.

Theorem 1.7. *An n-dimensional smooth manifold M is orientable if and only if the determinant line bundle $\Lambda^n TM$ admits a nonvanishing smooth section.*

Proof. ($\Rightarrow$) The transition maps of an oriented atlas preserve orientation. Thus, for charts $(U_\alpha, \varphi_\alpha)$ and (U_β, φ_β) with coordinates given by $\varphi_\alpha = (x^1, \dots, x^n)$ and $\varphi_\beta = (y^1, \dots, y^n)$

$$g_{\beta\alpha} := \det\left(\frac{\partial \varphi_\beta}{\partial \varphi_\alpha}\right) > 0, \qquad D(\varphi_\beta \circ \varphi_\alpha^{-1}) = \frac{\partial \varphi_\beta}{\partial \varphi_\alpha} := \left[\frac{\partial y^i}{\partial x^j}\right]_{1 \le i,j \le n}$$

On each local chart $(U_\alpha, \varphi_\alpha)$, define the nonvanishing local section of $\Lambda^n TM|_{U_\alpha}$ by

$$e_\alpha(p) = \frac{\partial}{\partial x^1}\Big|_p \wedge \cdots \wedge \frac{\partial}{\partial x^n}\Big|_p$$

and by the change-of-coordinates formula for the coordinate bases on $U_\alpha \cap U_\beta$, the positively oriented frames on U_α and U_β satisfy the compatibility condition $e_\alpha = g_{\beta\alpha} e_\beta$, which means e_α and e_β only differ by a positive function on $U_\alpha \cap U_\beta$ because the transition map has positive Jacobian determinant. Take a partition of unity $\{\rho_\alpha\}$ corresponding to the given oriented atlas such that $\operatorname{supp}(\rho_\alpha) \subseteq U_\alpha$ and let

$$s(p) = \sum_\alpha \rho_\alpha(p) e_\alpha(p) = \Big(\sum_\alpha \rho_\alpha(p) g_{\beta\alpha}(p)\Big) e_\beta(p) \quad \forall p \in M$$

This defines a global smooth section by extending the terms $\rho_\alpha e_\alpha$ by zero outside U_α. We may check that s is nowhere vanishing. On each chart, $e_p = \partial_1 \wedge \cdots \wedge \partial_n$ is nonvanishing. The partition of unity and the positivity of $g_{\alpha\beta}$ imply that its coefficient in s is strictly positive. Thus s is a smooth nonvanishing section.
($\Leftarrow$) Suppose there is a nonvanishing section $s \in \Gamma(\Lambda^n TM)$. On any connected chart U_α, write $s(p) = f_\alpha(p) e_\alpha(p)$ for some $f_\alpha \in C^\infty(U_\alpha)$, since the fiber $\Lambda^n T_p M$ is one-dimensional and is spanned by $e_\alpha(p)$. By replacing e_β with $-e_\beta$ when $f_\beta < 0$, one can always consider the case $f_\alpha(p) > 0\ \forall \alpha$, and indeed the nonvanishing section s gives an orientation on $T_p M$ for any $p \in M$ by declaring that $(v_1, \dots, v_n)$ is positively oriented if there exists $\lambda > 0$ such that

$$v_1 \wedge \cdots \wedge v_n = \lambda s(p)$$

Since the tangent bundle always trivializes on local charts $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in A}$, the orientation above can be defined on each chart. It remains to show that the positive charts are mutually compatible. Let $s = f_\alpha e_\alpha$ for some $f_\alpha \in C^\infty(U_\alpha)^\times$ be the restriction of the nonvanishing section to a chart. The compatibility condition of the top-degree exterior algebra on the tangent bundle is given by

$$f_\beta(p) = \det\left(\frac{\partial \varphi_\beta}{\partial \varphi_\alpha}\right) f_\alpha, \quad \forall p \in U_\alpha \cap U_\beta$$

since $e_\beta = \det(\partial \varphi_\alpha / \partial \varphi_\beta) e_\alpha$. Since s is nonvanishing, each coefficient function is nonvanishing and has constant sign on a connected chart. We may choose all charts to be connected. With the chosen positive frames, we obtain

$$\det D(\varphi_\beta \circ \varphi_\alpha^{-1}) = \det\left(\frac{\partial \varphi_\beta}{\partial \varphi_\alpha}\right) = \frac{f_\beta}{f_\alpha} > 0,$$

for any $\alpha, \beta \in A$ such that $U_\alpha \cap U_\beta \neq \emptyset$, which shows that the atlas $\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in A}$ is oriented. □

As a result of the determinant line bundle definition of orientation on manifolds, we may remark that in most cases when studying smooth manifolds, the language of differential forms is more convenient. Due to the canonical isomorphism $(\Lambda^k TM)^* \cong \Lambda^k T^* M$, one can safely use an isomorphism (not necessarily canonical) between the tangent and cotangent bundles to translate the orientation on the manifold to the language of n-forms: Recall that $\Omega^k(M) := \Gamma(\Lambda^k T^* M)$, and let $n := \dim M$.

Definition 1.8 (Volume Forms)**.** A volume form on M is an n-form $\omega \in \Omega^n(M)$ such that $\forall p \in M$, $\omega_p \in \Lambda^n T_p^* M$ is a nonzero form.

This gives an equivalent characterization of orientability in terms of differential forms.

Theorem 1.9. *M is orientable if and only if M admits a volume form.*

To prove this theorem, consider a nowhere-vanishing section $e \in \Gamma(\Lambda^n TM)$. Since $\Lambda^n T_p M$ is one-dimensional, define the dual section $\omega \in \Omega^n(M)$ pointwise by

$$\omega_p(e_p) = 1 \quad \forall p \in M$$

The smoothness of ω follows from that of e.

2 Normal Bundle and Framing

We begin with a special case: the orientation induced on the boundary of an oriented manifold.
Recall that a manifold with boundary M has an atlas $\{(U_\alpha, \varphi_\alpha)\}$ modeled on the upper half-space, $\varphi : U \subseteq M \xrightarrow{\cong} V \subseteq \mathbb{H}^n$, and the boundary of M is defined by

$$\partial M = \{p \in M \mid \exists \alpha \in A : \varphi_\alpha(p) \in \partial \mathbb{H}^n\}$$

In a boundary chart $\varphi_\alpha = (x^1, \dots, x^n)$ with $x^1 \geq 0$, boundary points satisfy $x^1(p) = 0$. For every $p \in \partial M$, the tangent space of the boundary is the subspace

$$T_p(\partial M) := \mathrm{Span}\{\partial_2, \dots, \partial_n\}$$

Thus, we may define the "normal" direction (more generally, the normal bundle) of the boundary as $\nu_{\partial M} := TM|_{\partial M} / T(\partial M) = \mathbb{R} \cdot \partial_1$ which makes the following sequence exact:

$$0 \longrightarrow T(\partial M) \xhookrightarrow{i} TM|_{\partial M} \twoheadrightarrow \nu_{\partial M} \longrightarrow 0$$

With this notion of normal direction, we call a vector $n_p \in \mathbb{R}_{<0} \cdot \partial_1|_p$ an outward-pointing normal vector (i.e., $(\mathrm{d}x^1)_p(n_p) < 0$). Note that the normal bundle $\nu_{\partial M}$ is always trivial.

Proposition 2.1. *The normal bundle of the boundary $\nu_{\partial M} \cong \partial M \times \mathbb{R}$ is a trivial bundle.*

We prove this fact after defining the induced boundary orientation.
We may define the induced orientation of the boundary as

Definition 2.2 (Induced Boundary Orientation)**.** Let M be an orientable smooth manifold with a chosen orientation, and let $n_p \in T_p M$ be an outward-pointing normal vector at $p \in \partial M$. Then, a frame $(v_1, \dots, v_{n-1}) \in \mathrm{Fr}_{n-1}\left(T_p(\partial M)\right)$ is positively oriented if and only if $(n_p, v_1, \dots, v_{n-1}) \in \mathrm{Fr}_n(T_p M)$ is positively oriented on M.

Because the definition uses boundary charts, we must check that it is independent of that choice.

Proposition 2.3. *The induced orientation on the boundary does not depend on the choice of boundary charts.*

Proof. Consider the transition map

$$\varphi_{\alpha\beta} := \varphi_\alpha \circ \varphi_\beta^{-1} = (\varphi^1_{\alpha\beta}, \dots, \varphi^n_{\alpha\beta})$$

between boundary charts U_α and U_β, and write $U_{\alpha\beta} := U_\alpha \cap U_\beta$. Along the boundary, $\varphi^1_{\alpha\beta}(0, x^2, \dots, x^n) = 0$, so

$$\frac{\partial \varphi_{\alpha\beta}}{\partial x^j} = 0, \quad \text{on } x^1 = 0,\ j = 2, \dots, n$$

Thus, at every $p \in \partial M$, the Jacobian has the block form

$$D(\varphi_{\alpha\beta})_p = \begin{pmatrix} A & 0 \\ * & c \end{pmatrix}$$

where

$$A := \left(\frac{\partial \varphi^i_{\alpha\beta}}{\partial x^j}\right)_{i,j=2,\dots,n}, \quad c = \frac{\partial \varphi^n_{\alpha\beta}}{\partial x^n} > 0$$

This implies $\det D(\varphi_{\alpha\beta})_p = c \det A$. Thus, if $D(\varphi_{\alpha\beta})_p > 0$, we have $\det A > 0$, and this is exactly the transition map of the tangent bundle of ∂M. Thus, the change of boundary chart preserves orientation. □

Furthermore, if $\omega \in \Omega^n(M)$ is the volume form corresponding to the chosen orientation on M, then the $(n-1)$-form corresponding to the induced orientation is $\eta := i_{n_{\text{out}}}(\omega|_{\partial M}) \in \Omega^{n-1}(\partial M)$, where n_{out} is an outward-pointing vector field. This follows by writing the forms in local coordinates.

Now we may prove Proposition 2.1 that the normal bundle of the boundary is trivial:

Proof of Proposition 2.1. It is enough to construct a smooth outward-pointing vector field along ∂M, since its image in

$$\nu_{\partial M} := TM|_{\partial M} / T\partial M$$

will be a nonvanishing smooth section.

Let $(U_\alpha, \varphi_\alpha)$ be boundary charts with

$$\varphi_\alpha(U_\alpha) \subset \mathbb{H}^n = \{x^1 \geq 0\}, \qquad B_\alpha := U_\alpha \cap \partial M.$$

On B_α, define

$$N_\alpha := f_\alpha \frac{\partial}{\partial x^1_\alpha}, \qquad f_\alpha \in C^\infty(U_\alpha, \mathbb{R}_{<0}).$$

Then $dx^1_\alpha(N_\alpha) = f_\alpha < 0$, so N_α is outward pointing.

We first note that the outward condition is independent of the chosen boundary chart. On overlaps, write $\varphi_{\alpha\beta} := \varphi_\alpha \circ \varphi_\beta^{-1} = (\varphi^1_{\alpha\beta}, \dots, \varphi^n_{\alpha\beta})$. Since $\varphi_{\alpha\beta}$ preserves the boundary and the interior side,

$$\varphi^1_{\alpha\beta}(0, x^2, \dots, x^n) = 0, \qquad x^1 > 0 \to \varphi^1_{\alpha\beta}(x) > 0.$$

Hence, along the boundary,

$$\mathrm{d}x^1_\alpha = \frac{\partial \varphi^1_{\alpha\beta}}{\partial x^1_\beta} \mathrm{d}x^1_\beta, \qquad \frac{\partial \varphi^1_{\alpha\beta}}{\partial x^1_\beta} > 0$$

Therefore, $dx^1_\alpha(v) < 0 \iff dx^1_\beta(v) < 0$. Thus, being outward pointing is coordinate-independent.

Now choose a smooth partition of unity $\{\rho_\alpha\}$ on ∂M subordinate to $\{B_\alpha\}$. Define

$$N := \sum_\alpha \rho_\alpha N_\alpha.$$

The sum is locally finite, so N is a smooth section of $TM|_{\partial M}$. For any $p \in \partial M$, choose a boundary chart $(U_\gamma, \varphi_\gamma)$ around p. Then

$$\mathrm{d}x^1_\gamma(N_p) = \sum_\alpha \rho_\alpha(p) \mathrm{d}x^1_\gamma(N_{\alpha,p}) < 0,$$

because $\rho_\alpha(p) \geq 0$, $\sum_\alpha \rho_\alpha(p) = 1$, and every $N_{\alpha,p}$ with $\rho_\alpha(p) \neq 0$ is outward pointing. Hence N is outward pointing everywhere on ∂M.

In particular, $N_p \neq 0 \ \forall p \in \partial M$. Moreover, since $T_p\partial M = \ker(dx^1_\gamma)_p$ and $(\mathrm{d}x^1_\gamma)_p(N_p) < 0$, we have $N_p \notin T_p\partial M$. Hence the projection $[N] \in \Gamma(\nu_{\partial M})$ is nowhere vanishing. □

We now extend the preceding construction from boundaries to general embedded submanifolds.

Definition 2.4 (Normal Bundle)**.** Let M be a smooth n-dimensional manifold and let $i : N^k \hookrightarrow M^n$ be an embedded submanifold. We regard TN as a subbundle of i^*TM via di. Then, the normal bundle of N in M is defined by

$$\nu_N(M) := i^*TM/TN$$

Furthermore, a framing of $N \subseteq M$ is a bundle isomorphism

$$\varphi : \nu_N(M) \xrightarrow{\cong} N \times \mathbb{R}^{n-k}$$

i.e., the submanifold N admits a framing if and only if its normal bundle is trivial.

A Riemannian metric on M gives an alternative description of the normal bundle as an orthogonal complement.

Definition 2.5 (Normal Bundle via Riemannian Metric)**.** Let (M, g) be a Riemannian manifold, and let $i : N^k \hookrightarrow M^n$ be an embedded submanifold. The normal bundle on N is given by

$$\nu_N(M, g) := \coprod_{p \in N} (T_pN)^{\perp_g} \subseteq i^*TM = TM|_N$$

where each fiber is the orthogonal complement of $i_{*,p}T_pN$ in $T_{i(p)}M$.

We show that these two definitions coincide.

Proposition 2.6. *Let (M, g) be a Riemannian manifold, and let $i : N^k \hookrightarrow M^n$ be an embedded submanifold. Then the natural quotient map*

$$q_g : \nu_N(M, g) \to \nu_N(M), \qquad v \mapsto [v] \mod T_pN$$

where $v \in (T_pN)^{\perp_g}$, is a vector bundle isomorphism.

Proof. For every $p \in N$, since g is positive definite on $T_{i(p)}M$, we have the orthogonal decomposition

$$T_{i(p)}M = i_{*,p}T_pN \oplus (T_pN)^{\perp_g}$$

Hence the induced fiberwise map $q_{g,p} : (T_pN)^{\perp_g} \to T_{i(p)}M/i_{*,p}T_pN$ is a linear isomorphism.
Indeed, if $v \in (T_pN)^{\perp_g}$ and $q_{g,p}(v) = 0$, then $v \in i_{*,p}T_pN$. Thus

$$v \in i_{*,p}T_pN \cap (T_pN)^{\perp_g}$$

so $g(v, v) = 0$, and therefore $v = 0$. Hence $q_{g,p}$ is injective. On the other hand, for any class $[w] \in T_{i(p)}M/i_{*,p}T_pN$, write $w = u + v$, where $u \in i_{*,p}T_pN$ and $v \in (T_pN)^{\perp_g}$. Then $[w] = [v]$. Hence $q_{g,p}$ is surjective.
It remains to check smoothness. Locally choose a frame $e_1, \dots, e_n$ of i^*TM such that $e_1, \dots, e_k$ is a frame of TN. Let

$$G = (g(e_a, e_b))_{1 \le a,b \le k}$$

Since $g|_{TN}$ is positive definite, G is invertible. For $\alpha = k+1, \dots, n$, define

$$\widetilde{e}_\alpha = e_\alpha - \sum_{a,b=1}^{k} (G^{-1})^{ab} g(e_\alpha, e_b) e_a$$

Then, $g(\widetilde{e}_\alpha, e_c) = 0$ for every $1 \le c \le k$. Hence, $\widetilde{e}_{k+1}, \dots, \widetilde{e}_n$ is a local frame of $\nu_N(M, g)$. Moreover, $q_g(\widetilde{e}_\alpha) = [e_\alpha]$. Since $[e_{k+1}], \dots, [e_n]$ is a local frame of $\nu_N(M)$, the local matrix of q_g is the identity matrix. Therefore, q_g is a smooth vector bundle isomorphism. □

As in the boundary case, the normal bundle fits into a short exact sequence:

Proposition 2.7. *The short exact sequence*

$$0 \longrightarrow TN \hookrightarrow i^*TM \twoheadrightarrow \nu_N(M) \longrightarrow 0$$

admits a metric-dependent splitting. Consequently, $i^*TM \cong TN \oplus \nu_N(M) \cong TN \oplus \nu_N(M, g)$.

Proof. Let $\pi : i^*TM \to \nu_N(M)$ be the quotient projection. By the previous proposition,

$$q_g = \pi|_{\nu_N(M,g)} : \nu_N(M, g) \to \nu_N(M)$$

is a vector bundle isomorphism. Hence

$$s_g := q_g^{-1} : \nu_N(M) \to \nu_N(M, g) \subseteq i^*TM$$

satisfies $\pi \circ s_g = \mathrm{id}_{\nu_N(M)}$. Thus, s_g is a splitting of the short exact sequence.
More explicitly, define $\Phi_g : TN \oplus \nu_N(M) \to i^*TM$ by $\Phi_g(u, \eta) = u + s_g(\eta)$. Its inverse is given by

$$\Phi_g^{-1}(w) = \big(w - s_g(\pi(w)), \pi(w)\big)$$

Indeed,

$$\pi\big(w - s_g(\pi(w))\big) = \pi(w) - \pi(s_g(\pi(w))) = \pi(w) - \pi(w) = 0$$

so $w - s_g(\pi(w)) \in \ker \pi = TN$. Therefore, $i^*TM \cong TN \oplus \nu_N(M)$. Since $q_g : \nu_N(M, g) \to \nu_N(M)$ is an isomorphism, we also obtain $i^*TM \cong TN \oplus \nu_N(M, g)$. □

As a direct corollary of Proposition 2.6, although we use the Riemannian metric to define the normal bundle for technical convenience, the nature of the normal bundle is topological.

Proposition 2.8. *Let* g_0, g_1 *be two Riemannian metrics on* M. *Let*

$$N_0 := \nu_N(M, g_0), \qquad N_1 := \nu_N(M, g_1)$$

be the corresponding orthogonal normal bundles. Then there exists a vector bundle isomorphism $N_0 \cong N_1$ *that maps the fiber at* p *to the fiber at* p.

The simplest way to prove this proposition is to consider the isomorphism $q_i : \nu_N(M, g_i) \to \nu_N(M)$. We give an alternative proof using a convex combination of Riemannian metrics.

Proof. Consider the convex combination of metrics $g_t = (1 - t)g_0 + tg_1$ for $t \in [0, 1]$. For every nonzero $v \in T_pM$, we have

$$g_t(v, v) = (1 - t)g_0(v, v) + tg_1(v, v) > 0$$

Hence g_t is a Riemannian metric for every $t \in [0, 1]$.
Let $N_t := \nu_N(M, g_t)$. By the previous proposition, for every $t \in [0, 1]$, the natural quotient map $q_t : N_t \to \nu_N(M)$ is a vector bundle isomorphism. Therefore we may define $A_t : N_0 \to N_t$ by $A_t = q_t^{-1} \circ q_0$. This gives a family of vector bundle isomorphisms covering id_N. Moreover, this family varies smoothly in t. Indeed, locally choose a frame $e_1, \ldots, e_n$ of i^*TM such that $e_1, \ldots, e_k$ is a frame of TN. Let

$$G_t = (g_t(e_a, e_b))_{1 \le a,b \le k}.$$

For $\alpha = k + 1, \ldots, n$, define

$$\widetilde{e}_\alpha(t) = e_\alpha - \sum_{a,b=1}^{k} (G_t^{-1})^{ab} g_t(e_\alpha, e_b) e_a$$

Then, $\widetilde{e}_{k+1}(t), \dots, \widetilde{e}_n(t)$ is a local frame of N_t, and it depends smoothly on t. If

$$v = \sum_{\alpha=k+1}^{n} \lambda_\alpha \widetilde{e}_\alpha(0) \in N_0$$

then

$$A_t(v) = \sum_{\alpha=k+1}^{n} \lambda_\alpha \widetilde{e}_\alpha(t)$$

In particular, taking $t = 1$, we get a vector bundle isomorphism $A_1 = q_1^{-1} \circ q_0 : N_0 \to N_1$. □

We conclude this section with a central application of the normal bundle.

Theorem 2.9 (Tubular Neighborhood Theorem)**.** *Let $i : N^k \hookrightarrow M^n$ be an embedded submanifold. Then there exists an open neighborhood $U \subseteq \nu_N(M)$ of the zero section and an open neighborhood $W \subseteq M$ of $i(N)$ such that there is a diffeomorphism $\Phi : U \xrightarrow{\cong} W$ satisfying $\Phi(0_p) = i(p)$ for every $p \in N$.*
Equivalently, after choosing a Riemannian metric g on M, there exists an open neighborhood $U_g \subseteq \nu_N(M, g)$ of the zero section such that the normal exponential map

$$\exp_g^\perp : U_g \to M, \qquad v \mapsto \exp^g_{i(p)}(v), \quad v \in (T_pN)^{\perp_g}$$

is a diffeomorphism from U_g onto an open neighborhood of $i(N)$ in M.

3 Proof of the Tubular Neighborhood Theorem

In this section, we prove the tubular neighborhood theorem by using the normal exponential map. Let $i : N^k \hookrightarrow M^n$ be an embedded submanifold. For simplicity, we identify N with its image $i(N)$, so that $T_pN \subseteq T_pM$ for every $p \in N$.
Choose a Riemannian metric g on M. Let

$$E := \nu_N(M, g) = \coprod_{p \in N} (T_pN)^{\perp_g}$$

be the orthogonal normal bundle, and let $\pi : E \to N$ be the bundle projection.

Definition 3.1 (Normal Exponential Map)**.** Let $\mathcal{D} \subseteq E$ be the set of all $v \in E$ such that the geodesic

$$t \mapsto \exp^g_{\pi(v)}(tv)$$

is defined at $t = 1$. Since geodesics exist locally and depend smoothly on initial conditions, $\mathcal{D}$ is an open neighborhood of the zero section of E.
The normal exponential map is defined by

$$\operatorname{Exp}_g^\perp : \mathcal{D} \to M, \qquad v \mapsto \exp^g_{\pi(v)}(v).$$

The first step is to compute the differential of the normal exponential map along the zero section.

Lemma 3.2. *For every $p \in N$, under the canonical identification*

$$T_{0_p}E \cong T_pN \oplus E_p,$$

and under the orthogonal decomposition $T_pM = T_pN \oplus E_p$, we have $\mathrm{d}(\operatorname{Exp}_g^\perp)_{0_p}(u, v) = u + v$. In particular, $\mathrm{d}(\operatorname{Exp}_g^\perp)_{0_p} : T_{0_p}E \to T_pM$ is a linear isomorphism.

Proof. Let $u \in T_pN$. Choose a smooth curve $\alpha : (-\epsilon, \epsilon) \to N$ such that $\alpha(0) = p$ and $\alpha'(0) = u$. Then $\mathrm{Exp}_g^\perp(0_{\alpha(t)}) = \alpha(t)$. Therefore

$$\mathrm{d}(\mathrm{Exp}_g^\perp)_{0_p}(u, 0) = \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} \mathrm{Exp}_g^\perp(0_{\alpha(t)}) = \alpha'(0) = u$$

Now let $v \in E_p = (T_pN)^{\perp_g}$. Consider the vertical curve $t \mapsto tv \in E_p$. Then, $\mathrm{Exp}_g^\perp(tv) = \exp_p^g(tv)$. Hence

$$\mathrm{d}(\mathrm{Exp}_g^\perp)_{0_p}(0, v) = \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} \exp_p^g(tv) = v$$

since the geodesic $t \mapsto \exp_p^g(tv)$ has initial velocity v.
By linearity,

$$\mathrm{d}(\mathrm{Exp}_g^\perp)_{0_p}(u, v) = u + v.$$

Since $T_pM = T_pN \oplus E_p$, this map is a linear isomorphism. □

Lemma 3.3 (Global Inverse Function Lemma Along the Zero Section)**.** *Let $E \to N$ be a vector bundle, and let $\mathcal{D} \subseteq E$ be an open neighborhood of the zero section. Suppose $F : \mathcal{D} \to M$ is a smooth map such that $F|_N : N \to M$ is an embedding and*

$$dF_{0_p} : T_{0_p}E \to T_{F(0_p)}M$$

is a linear isomorphism for every $p \in N$. Then there exists an open neighborhood $U \subseteq \mathcal{D}$ of the zero section such that

$$F|_U : U \to F(U)$$

is a diffeomorphism onto an open neighborhood of $F(N)$ in M.

Proof. By the inverse function theorem, for every $p \in N$, there exists an open neighborhood $\Omega_p \subseteq \mathcal{D}$ of 0_p such that

$$F|_{\Omega_p} : \Omega_p \to F(\Omega_p)$$

is a diffeomorphism onto an open subset of M.
Thus, after replacing $\mathcal{D}$ by a smaller open neighborhood of the zero section, we may assume that F is a local diffeomorphism on $\mathcal{D}$.
It remains to shrink $\mathcal{D}$ so that F becomes injective. Define

$$R = \{(x, y) \in \mathcal{D} \times \mathcal{D} : F(x) = F(y)\}.$$

Since M is Hausdorff, the diagonal $\Delta_M \subseteq M \times M$ is closed, and therefore $R = (F \times F)^{-1}(\Delta_M)$ is closed in $\mathcal{D} \times \mathcal{D}$.
Since F is a local diffeomorphism, for every $x \in \mathcal{D}$ there exists an open neighborhood $\Omega_x \subseteq \mathcal{D}$ such that $F|_{\Omega_x}$ is injective. Hence

$$\Omega := \bigcup_{x \in \mathcal{D}} \Omega_x \times \Omega_x$$

is an open neighborhood of the diagonal $\Delta_{\mathcal{D}}$, and $R \cap \Omega = \Delta_{\mathcal{D}} \cap \Omega$.
Since $F|_N$ is an embedding, it is injective. Hence

$$R \cap (N \times N) = \Delta_N.$$

Therefore, the closed set

$$C := R \cap \big((\mathcal{D} \times \mathcal{D}) \setminus \Omega\big)$$

is disjoint from $N \times N$.

By a standard shrinking argument for manifolds, we may choose an open neighborhood $U \subseteq \mathcal{D}$ of the zero section such that

$$(U \times U) \cap C = \varnothing$$

Now suppose $x, y \in U$ and $F(x) = F(y)$. Then $(x, y) \in R$. Since $(x, y) \notin C$, we must have $(x, y) \in \Omega$. Hence $(x, y) \in \Omega_z \times \Omega_z$ for some $z \in \mathcal{D}$, and $F|_{\Omega_z}$ is injective. Therefore, $x = y$. Thus $F|_U$ is injective.
Since F is a local diffeomorphism on U, the restriction

$$F|_U : U \to F(U)$$

is a diffeomorphism. Moreover, $F(U)$ is open in M, and since U contains the zero section, $F(U)$ is an open neighborhood of $F(N)$. □

Theorem 3.4 (Tubular Neighborhood Theorem)**.** *Let $i : N^k \hookrightarrow M^n$ be an embedded submanifold. Then there exists an open neighborhood $U \subseteq \nu_N(M)$ of the zero section and an open neighborhood $W \subseteq M$ of $i(N)$ such that there is a diffeomorphism*

$$\Phi : U \xrightarrow{\cong} W$$

satisfying $\Phi(0_p) = i(p)$ for every $p \in N$.

Proof. Choose a Riemannian metric g on M. By the previous discussion, the orthogonal normal bundle $E = \nu_N(M, g)$ is canonically isomorphic to the quotient normal bundle

$$\nu_N(M) = i^*TM/TN$$

via the natural quotient map

$$q_g : E \to \nu_N(M), \qquad v \mapsto [v] \mod T_pN$$

Consider the normal exponential map

$$\mathrm{Exp}_g^\perp : \mathcal{D} \to M, \qquad v \mapsto \exp^g_{\pi(v)}(v)$$

where $\mathcal{D} \subseteq E$ is an open neighborhood of the zero section.
For every $p \in N$, the previous lemma gives

$$\mathrm{d}(\mathrm{Exp}_g^\perp)_{0_p}(u, v) = u + v.$$

Since $T_pM = T_pN \oplus E_p$, this differential is a linear isomorphism. Also, $\mathrm{Exp}_g^\perp(0_p) = p$. Hence, the restriction of $\mathrm{Exp}_g^\perp$ to the zero section is precisely the embedding $i : N \hookrightarrow M$.
Therefore, the global inverse function lemma along the zero section applies to $F = \mathrm{Exp}_g^\perp$. Thus, there exists an open neighborhood $U_g \subseteq E = \nu_N(M, g)$ of the zero section such that

$$\mathrm{Exp}_g^\perp|_{U_g} : U_g \to \mathrm{Exp}_g^\perp(U_g)$$

is a diffeomorphism onto an open neighborhood of N in M.
Let $W := \mathrm{Exp}_g^\perp(U_g)$. Using the bundle isomorphism

$$q_g : \nu_N(M, g) \xrightarrow{\cong} \nu_N(M),$$

set

$$U := q_g(U_g) \subseteq \nu_N(M).$$

Define $\Phi : U \to W$ by $\Phi(\eta) = \mathrm{Exp}_g^\perp\big(q_g^{-1}(\eta)\big)$. Then Φ is a diffeomorphism. Moreover, for every $p \in N$,

$$\Phi(0_p) = \mathrm{Exp}_g^\perp(0_p) = i(p).$$

Therefore, U is a tubular neighborhood of the zero section in $\nu_N(M)$, and W is an open neighborhood of $i(N)$ in M diffeomorphic to it. □

Remark. After choosing a Riemannian metric g on M, the tubular neighborhood diffeomorphism may be written more concretely as

$$\mathrm{Exp}_g^{\perp} : U_g \subseteq \nu_N(M,g) \xrightarrow{\cong} W \subseteq M.$$

Thus, near N, the manifold M is diffeomorphic to an open neighborhood of the zero section in the normal bundle of N.

4 Regular Values and Preimages

The next bridge between differential topology and homotopy theory is the regular value theorem. It explains how a smooth map produces a submanifold, and in the case of maps to spheres, how a regular preimage naturally carries a framing.

Recall that $q \in N$ is a regular value of a smooth map $f \in C^\infty(M,N)$ if every point in $f^{-1}(q)$ is regular, that is, if $(\mathrm{d}f)_p : T_pM \to T_qN$ is surjective for every $p \in f^{-1}(q)$.

Definition 4.1 (Regular and Critical Points)**.** For a smooth map $f : M \to N$, $p \in M$ is a regular point if $f_{*,p}$ is surjective. The point $q \in N$ is said to be a regular value if $f^{-1}(q)$ contains only regular points. Otherwise, p is a critical point and $f(p)$ is a critical value.

We will repeatedly use the fact that the preimage of a regular value is an embedded submanifold of the domain.

Theorem 4.2 (Preimage of a Regular Value)**.** *Let $f : M^m \to N^n$ be a smooth map, and let $q \in N$ be a regular value. Then:*

- *$f^{-1}(q)$ is an $(m-n)$-dimensional embedded submanifold of M.*
- *$\forall p \in f^{-1}(q)$, the tangent space satisfies $T_p f^{-1}(q) = \ker(\mathrm{d}f)_p$.*

Proof. Choose charts (U,φ) around $p \in f^{-1}(q)$ and (V,ψ) around q such that $\varphi(p) = \psi(q) = 0$,

$$\tilde{f} := \psi \circ f \circ \varphi^{-1} : \varphi(U) \to \mathbb{R}^n, \quad \tilde{f}(0) = 0$$

q is a regular value of $\tilde{f}$ if and only if the Jacobian matrix $J(\tilde{f})$ has rank n at $0 \in \mathbb{R}^m$. After permuting coordinates, write $\tilde{f} = (f_1, \ldots, f_n)$ with $f_i : \varphi(U) \to \mathbb{R}^n$

$$\left| \frac{\partial(f_1,\ldots,f_n)}{\partial(x^1,\ldots,x^n)}(0) \right| \neq 0$$

To apply the inverse function theorem, define $F : \varphi(U) \to \mathbb{R}^m$ by

$$F(x^1,\ldots,x^m) := (f_1(x),\ldots,f_n(x), x^{n+1},\ldots,x^m)$$

This gives a local diffeomorphism $\varphi(U) \supseteq W \cong W'$. Define a new chart $(U' := \varphi^{-1}(W), \phi)$ with coordinates $\phi = (y^1,\ldots,y^m)$ by

$$\phi := (F|_W)^{-1} \circ \varphi : U' \to W', \text{ given explicitly by } \begin{cases} (y^1(x),\ldots,y^n(x)) = \tilde{f}(x) \\ (y^{n+1}(x),\ldots,y^m(x)) = (x^{n+1},\ldots,x^m) \end{cases}$$

where $f^{-1}(q) \cap U' \cong \tilde{f}^{-1}(0) \cap \psi(U')$ (since $\psi(q) = 0$) has coordinate representation

$$f^{-1}(q) \cap U' := \{p \in U' \mid y^1(p) = \cdots = y^n(p) = 0\}$$

By the local coordinate characterization of embedded submanifolds, this shows that $f^{-1}(q)$ is an $(m-n)$-dimensional submanifold.

Furthermore, $f^{-1}(q)$ clearly has tangent space given by

$$T_p f^{-1}(q) := \text{Span}\left\{ \frac{\partial}{\partial y^{n+1}}\Big|_p, \dots, \frac{\partial}{\partial y^m}\Big|_p \right\}$$

In these local coordinates, $F(y) = (y^1, \dots, y^n)$,

$$\ker(\mathrm{d}f)_p \cong \ker(JF) = \{v \in \mathbb{R}^m \mid v^1 = \cdots = v^n = 0\}$$

which shows that $\ker(\mathrm{d}f)_p = \text{Span}\{\partial_{y^{n+1}}, \dots, \partial_{y^m}\} = T_p f^{-1}(q)$. □

The rank of a smooth map $f : M \to N$ at $p \in M$ is the rank of the differential $(\mathrm{d}f)_p : T_pM \to T_{f(p)}N$. We recall the following terminology.

Definition 4.3 (Immersion, Embedding, and Submersion)**.** Let $f : M^m \to N^n$ be a C^k-map between differentiable manifolds. If $\text{rank}_p f = m \ \forall p \in M$, f is said to be a C^k-immersion. If $f : M \to f(M)$ is a C^k-immersion and also a homeomorphism ($f(M) \subseteq N$ has the subspace topology of N), then f is a C^k-embedding. If $\text{rank}_p f = n \ \forall p \in M$, then f is a C^k-submersion.

Equivalently, immersions and submersions are characterized by their differentials.

Proposition 4.4. *Let M and N be smooth manifolds. The smooth map $f : M \to N$ is said to be an immersion if $\forall p \in M : (\mathrm{d}f)_p : T_pM \to T_{f(p)}N$ is injective, and a submersion if $\forall p \in M : (\mathrm{d}f)_p : T_pM \to T_{f(p)}N$ is surjective (i.e., every point $p \in M$ is a regular point).*

This follows directly from the definitions.

A first observation that we can make from Theorem 4.2 is that when $\dim M = \dim N$, the "number of preimages" $|f^{-1}(q)|$ of a regular value is locally constant.

Proposition 4.5. *Let M and N be smooth manifolds with the same dimension, $f : M \to N$ be a smooth map with regular value $q \in N$, and M compact. Then, there exists a neighborhood $V \subseteq N$ of q such that $\forall q' \in V$*

$$|f^{-1}(q')| = |f^{-1}(q)|$$

Proof. By the inverse function theorem, $\forall p_i \in f^{-1}(q)$, there exists an open neighborhood U_i such that $f|_{U_i} : U_i \to V_i \subseteq N$ is a diffeomorphism. Then, we need the following lemma:

Lemma 4.6. *Let M and N be smooth manifolds with the same dimension. For M compact and $\forall f \in C^\infty(M, N)$, if $q \in N$ is a regular value, then*

$$f^{-1}(q) = \{p_1, \dots, p_k\}$$

is a finite subset of M.

To prove the lemma, note that $\dim f^{-1}(q) = 0$, so $f^{-1}(q)$ is discrete. It is also closed because $\{q\}$ is closed in N. Since M is compact, $f^{-1}(q)$ is compact and discrete, hence finite.

Since M is Hausdorff, we may choose $U_1, \dots, U_k$ to be pairwise disjoint and diffeomorphic to $V_1, \dots, V_k$. Then, in the open set

$$V = \Big(\bigcap_{i=1}^{k} V_i\Big) \setminus f\Big(M \setminus \bigcup_{i=1}^{k} U_i\Big)$$

Every point $q' \in V$ has exactly k preimages, so $|f^{-1}(q')| = k$. □

For later use, we restate the finiteness of a regular fiber when the domain is compact and the dimensions agree.

Proposition 4.7. *Let $f : M \to N$ be a smooth map between smooth manifolds. Suppose that M is compact and $\dim M = \dim N$. If $y \in N$ is a regular value of f, then $f^{-1}(y)$ is a finite set.*

Proof. Since y is a regular value of f, the regular value theorem implies that $f^{-1}(y)$ is a smooth submanifold of M of dimension

$$\dim f^{-1}(y) = \dim M - \dim N.$$

Because $\dim M = \dim N$, we have $\dim f^{-1}(y) = 0$. Thus $f^{-1}(y)$ is a 0-dimensional smooth manifold, hence a discrete topological space.

On the other hand, since f is continuous and $\{y\} \subset N$ is closed, the set $f^{-1}(y)$ is closed in M. Since M is compact, it follows that $f^{-1}(y)$ is compact.

Therefore, $f^{-1}(y)$ is both compact and discrete. A compact discrete space must be finite. Indeed, if $f^{-1}(y)$ were infinite, then the open cover

$$\big\{\{x\} : x \in f^{-1}(y)\big\}$$

would have no finite subcover, contradicting compactness. Hence $f^{-1}(y)$ is finite. □

When $\dim M \neq \dim N$, a regular fiber need not be a nonempty finite set. Sard's theorem instead concerns the size of the set of critical values in the target.

Theorem 4.8 (Sard's Theorem)**.** *For any $f \in C^\infty(M, N)$, the set of critical values of f has measure zero in N.*

A proof of Sard's theorem can be found in [1]; we omit it here.

Finally, Theorem 4.2 shows that, for a regular value q of $f : M \to N$ and every $p \in f^{-1}(q)$, the sequence

$$0 \longrightarrow T_p f^{-1}(q) \hookrightarrow T_p M \xrightarrow{(\mathrm{d}f)_p} T_q N \longrightarrow 0$$

is a short exact sequence. Thus, the sequence above leads to the framing of the preimage $f^{-1}(q)$ since the image $q \in N$ is a fixed point, i.e., $\forall p \in f^{-1}(q)$, $\nu_{f^{-1}(q)}(M)_p \cong T_q N$, and these fiberwise isomorphisms assemble into a smooth vector bundle isomorphism:

$$\nu_{f^{-1}(q)}(M) \cong f^{-1}(q) \times T_q N$$

which leads to the framing on $f^{-1}(q)$ in M.

5 Framed Cobordism

The previous section shows that the preimage of a regular value is not merely an embedded submanifold. More importantly, it has a canonical normal bundle description. In particular, if $f : M^m \to N^r$ is a smooth map and $q \in N$ is a regular value, then

$$\nu_{f^{-1}(q)}(M) \cong f^{-1}(q) \times T_q N.$$

Thus, after choosing a basis of the fixed vector space $T_q N$, the regular preimage $f^{-1}(q)$ becomes a framed submanifold of M. The purpose of this section is to define the equivalence relation on such framed submanifolds that corresponds to the homotopy of maps.

Throughout this section, M^m is a smooth manifold without boundary unless otherwise stated.

Definition 5.1 (Framed Submanifold)**.** Let $X^{m-r} \subseteq M^m$ be a compact embedded submanifold of codimension r. The normal bundle of X in M is defined by

$$\nu_X(M) := TM|_X / TX.$$

A framing of X in M is a vector bundle isomorphism $\varphi : \nu_X(M) \xrightarrow{\cong} X \times \mathbb{R}^r$. A framed submanifold of M is a pair (X, φ), where $X \subseteq M$ is a compact embedded submanifold and φ is a framing of its normal bundle.

Equivalently, after choosing a Riemannian metric on M and identifying $\nu_X(M) \cong TX^\perp$, a framing may be represented by r smooth normal vector fields $(v_1, \dots, v_r) \in \Gamma(TX^\perp)^{\oplus r}$ such that for every $x \in X$, the ordered tuple $(v_1(x), \dots, v_r(x))$ is a basis of the normal space $\nu_X(M)_x$. Hence, a framing is stronger than an orientation of the normal bundle: it trivializes the whole normal bundle, not just its determinant line.

We also allow the empty submanifold $\emptyset \subseteq M$. It has a unique framing and will be regarded as a framed submanifold.

Proposition 5.2. *Let M be an oriented smooth manifold and let (X, φ) be a framed submanifold of M. Then the framing φ induces an orientation on X.*

Proof. Since $X \subseteq M$ is an embedded submanifold, we have a short exact sequence of vector bundles over X:

$$0 \longrightarrow TX \hookrightarrow TM|_X \twoheadrightarrow \nu_X(M) \longrightarrow 0.$$

Taking determinant lines gives the canonical isomorphism

$$\det(TM|_X) \cong \det(TX) \otimes \det(\nu_X(M)).$$

The orientation of M gives an orientation of $\det(TM|_X)$, while the framing

$$\varphi : \nu_X(M) \xrightarrow{\cong} X \times \mathbb{R}^r$$

gives an orientation of $\nu_X(M)$ by pulling back the standard orientation of $\mathbb{R}^r$. Therefore, the determinant-line isomorphism above uniquely determines an orientation of $\det(TX)$, and hence an orientation of X. □

The main equivalence relation is the following.

Definition 5.3 (Framed Cobordism)**.** Let (X_0, φ_0) and (X_1, φ_1) be compact framed submanifolds of M^m of codimension r. We say that (X_0, φ_0) and (X_1, φ_1) are framed cobordant if there exists a compact, properly embedded submanifold with boundary

$$W^{m-r+1} \subseteq M \times [0,1]$$

such that $\partial W = X_0 \times \{0\} \sqcup X_1 \times \{1\}$ as a set. If orientations are included, this identity is understood as

$$\partial W = (-X_0) \times \{0\} \sqcup X_1 \times \{1\}$$

Moreover, W is required to be cylindrical near the boundary, i.e., there exists $\varepsilon > 0$ such that

$$W \cap (M \times [0, \varepsilon)) = X_0 \times [0, \varepsilon) \quad \text{and} \quad W \cap (M \times (1-\varepsilon, 1]) = X_1 \times (1-\varepsilon, 1]$$

Finally, W is equipped with a framing $\Phi : \nu_W(M \times [0,1]) \xrightarrow{\cong} W \times \mathbb{R}^r$ whose restrictions to the boundary agree with φ_0 and φ_1.

Let us explain the last condition more explicitly. Near the boundary, W is a product. Hence, along $X_i \times \{i\}$, $i = 0, 1$, we have

$$TW|_{X_i \times \{i\}} = TX_i \oplus \mathbb{R}\partial_t$$

while

$$T(M \times [0,1])|_{X_i \times \{i\}} = TM|_{X_i} \oplus \mathbb{R}\partial_t$$

Therefore,

$$\begin{aligned} \nu_W(M \times [0,1])|_{X_i \times \{i\}} &\cong \frac{TM|_{X_i} \oplus \mathbb{R}\partial_t}{TX_i \oplus \mathbb{R}\partial_t} \cong \frac{TM|_{X_i}}{TX_i} \\ &= \nu_{X_i}(M) \end{aligned}$$

Under this canonical identification, the restriction of Φ to $X_i \times \{i\}$ is required to be φ_i. In other words,

$$\Phi|_{X_0 \times \{0\}} = \varphi_0, \qquad \Phi|_{X_1 \times \{1\}} = \varphi_1$$

If such a framed cobordism exists, we write $(X_0, \varphi_0) \sim (X_1, \varphi_1)$.

Proposition 5.4. *Framed cobordism is an equivalence relation.*

Proof. We verify reflexivity, symmetry, and transitivity.
First, reflexivity follows from the product cobordism. For a framed submanifold (X, φ), take $W := X \times [0,1] \subseteq M \times [0,1]$. Then,

$$\partial W = X \times \{0\} \sqcup X \times \{1\}$$

and W is cylindrical near the boundary. Moreover,

$$\nu_{X\times[0,1]}(M \times [0,1]) \cong \pi_X^* \nu_X(M),$$

where $\pi_X : X \times [0,1] \to X$ is the projection. Hence, the framing φ pulls back to a framing $\Phi := \pi_X^* \varphi$ of W. Therefore, $(X, \varphi) \sim (X, \varphi)$.
Second, suppose $W \subseteq M \times [0,1]$ is a framed cobordism from (X_0, φ_0) to (X_1, φ_1). Consider the diffeomorphism

$$R : M \times [0,1] \to M \times [0,1], \qquad R(x,t) = (x, 1-t).$$

Then $R(W)$ is a cobordism from X_1 to X_0. The framing of W is carried forward by R to a framing of $R(W)$. Thus, $(X_1, \varphi_1) \sim (X_0, \varphi_0)$.
Finally, suppose

$$(X_0, \varphi_0) \sim (X_1, \varphi_1), \qquad (X_1, \varphi_1) \sim (X_2, \varphi_2)$$

Let W_{01} be a framed cobordism from (X_0, φ_0) to (X_1, φ_1) and let W_{12} be a framed cobordism from (X_1, φ_1) to (X_2, φ_2). After rescaling the interval, we may regard W_{01} as lying in $M \times [0, \frac{1}{2}]$ and W_{12} as lying in $M \times [\frac{1}{2}, 1]$. Since both cobordisms are cylindrical near X_1, the union

$$W_{02} := W_{01} \cup_{X_1} W_{12}$$

is a smooth properly embedded submanifold of $M \times [0,1]$. The framings also glue smoothly because both of them restrict to the same framing φ_1 on X_1. Therefore, W_{02} is a framed cobordism from (X_0, φ_0) to (X_2, φ_2). Hence, $(X_0, \varphi_0) \sim (X_2, \varphi_2)$. This proves that framed cobordism is an equivalence relation. □

Definition 5.5 (Framed Cobordism Classes)**.** The set of framed cobordism classes of compact codimension r framed submanifolds of M^m is denoted by

$$\Omega^{\mathrm{fr}}_{m-r}(M)$$

Here, the subscript records the dimension of the framed submanifold. The codimension r is kept implicit in the notation.

At this point, $\Omega^{\mathrm{fr}}_{m-r}(M)$ should be regarded as a set of equivalence classes. In special situations, such as $M = S^m$, this set will later be identified with a homotopy set

$$[S^m, S^r] = \pi_m(S^r)$$

which carries the usual homotopy group structure when $m \geq 1$ and $r \geq 1$.
We now explain how framed submanifolds arise naturally from regular preimages. The following proposition has been informally stated at the end of the last section:

Proposition 5.6 (Regular Preimages Are Framed)**.** *Let $f : M^m \to N^r$ be a smooth map and let $q \in N$ be a regular value of f. Then $X := f^{-1}(q)$ is a codimension r embedded submanifold of M. Moreover, after choosing an ordered basis $\beta = (e_1, \ldots, e_r)$ of T_qN, the submanifold X has a canonical framing*

$$\varphi_{f,q,\beta} : \nu_X(M) \xrightarrow{\cong} X \times \mathbb{R}^r$$

Proof. By Theorem 4.2, since q is a regular value, $X = f^{-1}(q)$ is a smooth embedded submanifold of dimension $m - r$, and $T_pX = \ker(\mathrm{d}f)_p$ for every $p \in X$. Hence, we have a short exact sequence

$$0 \longrightarrow T_pX \hookrightarrow T_pM \xrightarrow{(\mathrm{d}f)_p} T_qN \longrightarrow 0$$

Taking the quotient by T_pX, the differential induces a linear isomorphism

$$\overline{(\mathrm{d}f)_p} : T_pM/T_pX \xrightarrow{\cong} T_qN$$

Since $T_pM/T_pX = \nu_X(M)_p$, this gives $\nu_X(M)_p \cong T_qN$ for every $p \in X$. Since q is fixed, these fiberwise isomorphisms assemble into a smooth vector bundle isomorphism $\nu_X(M) \cong X \times T_qN$.
Finally, the chosen basis $\beta = (e_1, \ldots, e_r)$ identifies $T_qN \cong \mathbb{R}^r$. Therefore,

$$\nu_X(M) \cong X \times T_qN \cong X \times \mathbb{R}^r$$

which is the required framing. □

Explicitly, the framing in Proposition 5.6 is given by

$$\varphi_{f,q,\beta}(p, [v]) = (p, [(\mathrm{d}f)_p(v)]_\beta),$$

where $[v] \in T_pM/T_pX$ and $[(\mathrm{d}f)_p(v)]_\beta$ denotes the coordinate vector of $(\mathrm{d}f)_p(v) \in T_qN$ with respect to the basis β.
For the Pontryagin–Thom construction, the most important case is $f : M^m \to S^r$. If $q \in S^r$ is a regular value, then the framed submanifold is $f^{-1}(q) \subseteq M$, with normal framing induced by the differential

$$(\mathrm{d}f)_p : T_pM \to T_qS^r.$$

We first recall relative transversality, then use it to show that homotopies give framed cobordisms.

Theorem 5.7 (Relative Transversality)**.** *Let A be a smooth manifold with boundary and let $g : A \to N$ be a smooth map. Suppose that g is already transverse to a submanifold $Y \subseteq N$ on a neighborhood of ∂A. Then g can be perturbed, while fixed on a neighborhood of ∂A, to a smooth map $\tilde{g} : A \to N$ such that $\tilde{g} \pitchfork Y$.*

We will use this theorem without proof. It is a standard relative version of the transversality theorem.

Proposition 5.8 (Homotopy Gives Framed Cobordism)**.** *Let M^m be compact. Let*

$$f_0, f_1 : M \to S^r$$

be smooth maps, and let $q \in S^r$ be a regular value of both f_0 and f_1. Suppose that f_0 and f_1 are smoothly homotopic. Then, after choosing a basis of T_qS^r, the framed submanifolds $f_0^{-1}(q)$, $f_1^{-1}(q)$ are framed cobordant.

Proof. Let $H : M \times [0,1] \to S^r$ be a smooth homotopy from f_0 to f_1, so that

$$H(-, 0) = f_0, \qquad H(-, 1) = f_1$$

First, choose a smooth function $\rho : [0,1] \to [0,1]$ such that

$$\rho(t) = 0 \quad \text{for } t \in [0, \varepsilon), \qquad \rho(t) = 1 \quad \text{for } t \in (1 - \varepsilon, 1]$$

for some sufficiently small $\varepsilon > 0$. Replacing H by

$$H'(x, t) := H(x, \rho(t))$$

we may assume that the homotopy is constant near $t = 0$ and $t = 1$.

Since q is a regular value of both f_0 and f_1, the map H' is already transverse to q near the boundary of $M \times [0,1]$. By Theorem 5.7, after a small perturbation fixed near the boundary, we may assume that $H' \pitchfork \{q\}$. For simplicity, we still denote the perturbed homotopy by H.

Define $W := H^{-1}(q) \subseteq M \times [0,1]$. Since H is transverse to q, the regular value theorem implies that W is a smooth embedded submanifold of codimension r. Since M is compact, W is compact. Moreover, because H is constant near $t = 0$ and $t = 1$, we have

$$W \cap (M \times [0,\varepsilon)) = f_0^{-1}(q) \times [0,\varepsilon) \quad \text{and} \quad W \cap (M \times (1-\varepsilon,1]) = f_1^{-1}(q) \times (1-\varepsilon,1]$$

Thus,

$$\partial W = f_0^{-1}(q) \times \{0\} \sqcup f_1^{-1}(q) \times \{1\}$$

It remains to construct the framing on W. For every $z = (x,t) \in W$, transversality gives a short exact sequence

$$0 \longrightarrow T_z W \hookrightarrow T_z(M \times [0,1]) \xrightarrow{(\mathrm{d}H)_z} T_q S^r \longrightarrow 0.$$

Hence, $\nu_W(M \times [0,1])_z \cong T_q S^r$. Since q is fixed, these fiberwise isomorphisms assemble into a vector bundle isomorphism

$$\nu_W(M \times [0,1]) \cong W \times T_q S^r$$

Choosing a basis of $T_q S^r$ gives a framing $\Phi : \nu_W(M \times [0,1]) \xrightarrow{\cong} W \times \mathbb{R}^r$.

On the boundary component $f_0^{-1}(q) \times \{0\}$, the differential $(\mathrm{d}H)$ agrees with $(\mathrm{d}f_0)$ in the M-direction. Therefore, the induced framing restricts to the regular preimage framing of $f_0^{-1}(q)$. Similarly, on the boundary component $f_1^{-1}(q) \times \{1\}$, it restricts to the regular preimage framing of $f_1^{-1}(q)$. Therefore, W is a framed cobordism from $f_0^{-1}(q)$ to $f_1^{-1}(q)$. □

Consequently, the framed cobordism class of a regular preimage depends only on the homotopy class of the original map. More precisely, if $f_0 \simeq f_1 : M \to S^r$, then

$$[f_0^{-1}(q)]_{\mathrm{fr}} = [f_1^{-1}(q)]_{\mathrm{fr}} \in \Omega^{\mathrm{fr}}_{m-r}(M)$$

provided q is chosen as a regular value of the relevant maps after a possible small perturbation.

The converse direction, namely the construction of a map from a framed submanifold, will be the content of the Pontryagin–Thom construction. The key geometric input is the tubular neighborhood theorem.

Proposition 5.9 (Framed Tubular Neighborhood)**.** *Let (X, φ) be a codimension r framed submanifold of M^m. Then there exists a tubular neighborhood U of X in M and a diffeomorphism*

$$\Theta : X \times D^r \xrightarrow{\cong} U$$

such that $\Theta(x,0) = x$ for every $x \in X$. Moreover, the derivative of Θ in the D^r-direction realizes the chosen framing φ.

Proof. By the tubular neighborhood theorem, there exists an open neighborhood U of X in M and a diffeomorphism from a neighborhood of the zero section in $\nu_X(M)$ onto U. Since X is framed, we have a vector bundle isomorphism

$$\varphi : \nu_X(M) \xrightarrow{\cong} X \times \mathbb{R}^r$$

Shrinking the tubular neighborhood if necessary, we may restrict to a disk bundle in $\nu_X(M)$ which corresponds under φ to $X \times D^r \subseteq X \times \mathbb{R}^r$. Composing the inverse of this trivialization with the tubular neighborhood diffeomorphism gives

$$\Theta : X \times D^r \xrightarrow{\cong} U$$

By construction, $\Theta(x,0) = x$, and the normal coordinate directions in D^r agree with the chosen framing. □

Similarly, if $(W, \Phi) \subseteq M \times [0,1]$ is a framed cobordism, then the tubular neighborhood theorem and the framing Φ give a product tubular neighborhood $W \times D^r \subseteq M \times [0,1]$.

This compatibility is the geometric reason that framed cobordism corresponds to homotopy. A framed submanifold produces a map by collapsing the complement of a framed tubular neighborhood, and a framed cobordism produces a homotopy between the corresponding collapse maps.

The next section will turn this geometric relation into the Pontryagin–Thom correspondence.

6 The Pontryagin–Thom Construction

The previous section introduced framed cobordism as the geometric equivalence relation on framed submanifolds. We now explain why this equivalence relation is naturally related to homotopy classes of maps to spheres.

The main idea is simple. If (X, φ) is a codimension r framed submanifold of M, then a framed tubular neighborhood identifies a neighborhood of X with $X \times D^r$. The Pontryagin–Thom construction sends a point in this tubular neighborhood to its normal coordinate in D^r, and sends every point outside the tubular neighborhood to the basepoint of

$$S^r \cong D^r / \partial D^r$$

Thus, a framed submanifold produces a map $M \to S^r$.

Conversely, if $f : M \to S^r$ is a smooth map and $q \in S^r$ is a regular value, then $f^{-1}(q)$ is a framed submanifold of M. These two constructions are inverse to each other up to framed cobordism and homotopy.

Throughout this section, let M^m be a compact smooth manifold without boundary. We write

$$S^r := D^r / \partial D^r$$

and denote the image of ∂D^r by $*$, the basepoint of S^r.

Definition 6.1 (Pontryagin–Thom Collapse Map)**.** Let (X, φ) be a compact codimension r framed submanifold of M^m. Choose a framed tubular neighborhood

$$\Theta : X \times D^r \xrightarrow{\cong} U \subseteq M$$

such that $\Theta(x, 0) = x$. The Pontryagin–Thom collapse map associated to (X, φ) is the continuous map

$$c_{X,\varphi} : M \to S^r = D^r / \partial D^r$$

defined by

$$c_{X,\varphi}(p) = \begin{cases} [v], & p = \Theta(x, v) \in X \times D^r, \\ *, & p \in M \setminus \mathring{U}. \end{cases}$$

Here $[v]$ denotes the image of $v \in D^r$ under the quotient map $D^r \to D^r / \partial D^r$. The definition is continuous because if p approaches the boundary of the tubular neighborhood, then the normal coordinate v approaches ∂D^r, whose image in the quotient is exactly the basepoint $*$.

In particular,

$$c_{X,\varphi}^{-1}(0) = X,$$

where $0 \in D^r \subseteq D^r / \partial D^r$ denotes the image of the origin. Thus, the framed submanifold X is recovered as the inverse image of the center of the disk.

Proposition 6.2. *Let (X, φ) be a codimension r framed submanifold of M. Then $0 \in D^r/\partial D^r$ is a regular value of the collapse map $c_{X,\varphi}$ after smoothing the collapse map near ∂U, and the regular preimage framing of*

$$c_{X,\varphi}^{-1}(0) = X$$

agrees with the original framing φ.

Proof. On the tubular neighborhood $U \cong X \times D^r$, the collapse map is given by projection to the normal coordinate:

$$c_{X,\varphi}(x, v) = [v].$$

Near the zero section, this is simply the projection $X \times D^r \to D^r$. Therefore, $c_{X,\varphi}^{-1}(0) = X \times \{0\} \cong X$.

At a point $(x, 0) \in X \times D^r$, the differential of the local projection is

$$T_x X \oplus T_0 D^r \to T_0 D^r, \qquad (\xi, \eta) \mapsto \eta.$$

Hence, it is surjective, and its kernel is exactly $T_x X$. Therefore, 0 is a regular value near X.

The induced normal bundle isomorphism is $\nu_X(M) \cong X \times T_0 D^r$. Under the identification $T_0 D^r \cong \mathbb{R}^r$, this is precisely the framing determined by the tubular neighborhood and the chosen trivialization

$$\varphi : \nu_X(M) \xrightarrow{\cong} X \times \mathbb{R}^r$$

Thus, the regular preimage framing agrees with φ.

The possible non-smoothness of $c_{X,\varphi}$ occurs only near ∂U, where the quotient collapses ∂D^r to the basepoint. One may replace the radial coordinate by a smooth bump function which is constant near ∂D^r. This smoothing does not affect the map near $X \times \{0\}$, and hence does not change the regular preimage or its framing. □

We next show that the homotopy class of the collapse map does not depend on the auxiliary choices.

Proposition 6.3. *The homotopy class of the Pontryagin–Thom collapse map*

$$[c_{X,\varphi}] \in [M, S^r]$$

depends only on the framed submanifold (X, φ), not on the chosen framed tubular neighborhood.

Proof. Suppose $\Theta_0 : X \times D^r \to U_0$ and $\Theta_1 : X \times D^r \to U_1$ are two framed tubular neighborhoods compatible with the same framing φ. Since both tubular neighborhoods realize the same normal bundle trivialization along the zero section, they are isotopic after possibly shrinking the disk radius. Thus, there exists a smooth family of framed tubular neighborhoods

$$\Theta_t : X \times D^r \to U_t, \qquad t \in [0, 1],$$

with Θ_0 and Θ_1 as endpoints, after replacing D^r by a smaller disk if necessary.

Applying Definition 6.1 to Θ_t gives a continuous family of maps

$$c_t : M \to S^r.$$

This family is a homotopy from $c_{X,\varphi}^{(0)}$ to $c_{X,\varphi}^{(1)}$. Hence the homotopy class of the collapse map is independent of the chosen framed tubular neighborhood. □

Therefore, a framed submanifold determines a well-defined homotopy class of maps to S^r.

Definition 6.4 (The Pontryagin–Thom Map)**.** Define

$$\mathrm{PT} : \Omega_{m-r}^{\mathrm{fr}}(M) \to [M, S^r]$$

by $\mathrm{PT}([X, \varphi]) := [c_{X,\varphi}]$, where $c_{X,\varphi} : M \to S^r$ is the Pontryagin–Thom collapse map.

We must check that this is well-defined on framed cobordism classes.

Proposition 6.5. *If two framed submanifolds (X_0, φ_0) and (X_1, φ_1) are framed cobordant in M, then their Pontryagin–Thom collapse maps are homotopic, i.e., $c_{X_0,\varphi_0} \simeq c_{X_1,\varphi_1}$.*

Proof. Let $W^{m-r+1} \subseteq M \times [0,1]$ be a framed cobordism from (X_0, φ_0) to (X_1, φ_1), with framing

$$\Phi : \nu_W(M \times [0,1]) \xrightarrow{\cong} W \times \mathbb{R}^r$$

By the framed tubular neighborhood theorem, there is a tubular neighborhood

$$\Theta : W \times D^r \xrightarrow{\cong} U_W \subseteq M \times [0,1]$$

compatible with the framing Φ.

Since W is cylindrical near the boundary, the tubular neighborhood can also be chosen cylindrical near the boundary. That is, for sufficiently small $\varepsilon > 0$, near $t = 0$ it agrees with $X_0 \times [0, \varepsilon) \times D^r$, and near $t = 1$ it agrees with

$$X_1 \times (1 - \varepsilon, 1] \times D^r.$$

Define a collapse map $C : M \times [0,1] \to S^r$ by

$$C(p,t) = \begin{cases} [v], & (p,t) = \Theta(w,v) \in W \times D^r, \\ *, & (p,t) \notin \mathring{U}_W \end{cases}$$

This is continuous for the same reason as in Definition 6.1. Since the tubular neighborhood is cylindrical near the boundary, the restriction of C to $M \times \{0\}$ is the collapse map associated to (X_0, φ_0), and the restriction of C to $M \times \{1\}$ is the collapse map associated to (X_1, φ_1). Therefore, C is a homotopy $c_{X_0,\varphi_0} \simeq c_{X_1,\varphi_1}$. □

Hence, the map $\mathrm{PT} : \Omega^{\mathrm{fr}}_{m-r}(M) \to [M, S^r]$ is well-defined.

Now we construct the inverse direction. Let $f : M \to S^r$ be a smooth map. By Sard's theorem, regular values exist densely in S^r. Choose a regular value $q \in S^r \setminus \{*\}$. Then $f^{-1}(q)$ is a codimension r submanifold of M. Since q is fixed, the differential of f gives a canonical normal bundle identification

$$\nu_{f^{-1}(q)}(M) \cong f^{-1}(q) \times T_q S^r.$$

After choosing a basis of $T_q S^r$, this becomes a framing.

Definition 6.6 (Regular Preimage Construction)**.** Let $f : M \to S^r$ be a smooth map and let $q \in S^r \setminus \{*\}$ be a regular value. Choose an ordered basis β of $T_q S^r$. Define

$$\mathrm{RP}(f) := \big(f^{-1}(q), \varphi_{f,q,\beta}\big),$$

where $\varphi_{f,q,\beta}$ is the framing induced by $(\mathrm{d}f)_p : T_pM \to T_qS^r$ for $p \in f^{-1}(q)$.

The notation RP stands for regular preimage.

Proposition 6.7. *The framed cobordism class of* $\mathrm{RP}(f)$ *depends only on the homotopy class of f.*

Proof. Suppose $f_0 \simeq f_1 : M \to S^r$. Choose a smooth homotopy $H : M \times [0,1] \to S^r$ from f_0 to f_1. After a perturbation relative to the boundary, we may assume that H is transverse to the chosen regular value q. Then, $W := H^{-1}(q)$ is a compact embedded submanifold of $M \times [0,1]$ with boundary $\partial W = f_0^{-1}(q) \times \{0\} \sqcup f_1^{-1}(q) \times \{1\}$.

Moreover, for every $z \in W$, transversality gives a short exact sequence

$$0 \longrightarrow T_zW \hookrightarrow T_z(M \times [0,1]) \xrightarrow{(\mathrm{d}H)_z} T_qS^r \longrightarrow 0$$

Hence, $\nu_W(M \times [0,1]) \cong W \times T_qS^r$. Choosing the same basis of T_qS^r gives a framing of W. Its boundary restrictions agree with the regular preimage framings of $f_0^{-1}(q)$ and $f_1^{-1}(q)$. Therefore, $\mathrm{RP}(f_0) \sim \mathrm{RP}(f_1)$. □

The construction is also independent of the chosen regular value, up to framed cobordism.

Proposition 6.8. *Let $q_0, q_1 \in S^r \setminus \{*\}$ be regular values of $f : M \to S^r$. After choosing compatible bases of $T_{q_0}S^r$ and $T_{q_1}S^r$, the framed submanifolds $f^{-1}(q_0)$ and $f^{-1}(q_1)$ are framed cobordant.*

Proof. Choose a diffeomorphism $a : S^r \to S^r$ isotopic to the identity such that $a(q_0) = q_1$. Then $a \circ f \simeq f$. Since q_1 is a regular value of $a \circ f$ precisely when q_0 is a regular value of f, we have

$$(a \circ f)^{-1}(q_1) = f^{-1}(q_0).$$

By Proposition 6.7, the regular preimage of $a \circ f$ at q_1 is framed cobordant to the regular preimage of f at q_1. Therefore, $f^{-1}(q_0) \sim f^{-1}(q_1)$. □

We now state the main correspondence.

Theorem 6.9 (Pontryagin–Thom Correspondence)**.** *Let M^m be a compact smooth manifold without boundary. Then the Pontryagin–Thom construction gives a natural bijection*

$$\mathrm{PT} : \Omega^{\mathrm{fr}}_{m-r}(M) \xrightarrow{\cong} [M, S^r].$$

Proof. We construct inverse maps.

First, starting from a framed submanifold (X, φ), Definition 6.1 gives the collapse map $c_{X,\varphi} : M \to S^r$. Thus, we have a map

$$\mathrm{PT} : \Omega^{\mathrm{fr}}_{m-r}(M) \to [M, S^r]$$

Conversely, starting from a homotopy class $[f] \in [M, S^r]$, choose a smooth representative $f : M \to S^r$ and a regular value $q \in S^r \setminus \{*\}$. The regular preimage construction gives a framed submanifold

$$\mathrm{RP}(f) = \left(f^{-1}(q), \varphi_{f,q}\right)$$

By Propositions 6.7 and 6.8, its framed cobordism class depends only on $[f]$. Hence, we obtain a well-defined map

$$\mathrm{RP} : [M, S^r] \to \Omega^{\mathrm{fr}}_{m-r}(M).$$

It remains to prove that PT and RP are inverse to each other.

First, let (X, φ) be a framed submanifold. Consider its collapse map $c_{X,\varphi} : M \to S^r$. By Proposition 6.2, $c^{-1}_{X,\varphi}(0) = X$, and the regular preimage framing agrees with the original framing φ. Therefore,

$$\mathrm{RP}(\mathrm{PT}([X, \varphi])) = [X, \varphi]$$

Second, let $f : M \to S^r$ be a smooth map, and let $q \in S^r \setminus \{*\}$ be a regular value. We need to compare f with the collapse map associated with the framed submanifold $X = f^{-1}(q)$.

Choose a small coordinate ball $B \subseteq S^r$ centered at q, and choose coordinates $B \cong D^r$ sending q to 0. Since q is a regular value, the map f is a submersion near X. Hence, after shrinking B, the preimage $f^{-1}(B)$ is a tubular neighborhood of X, and the map f on this neighborhood is identified with the normal coordinate projection $X \times D^r \to D^r$.

Now choose a map $h : S^r \to S^r$ which is homotopic to the identity, sends $S^r \setminus B$ to the basepoint $*$, and agrees with the quotient projection

$$D^r \to D^r/\partial D^r$$

on $B \cong D^r$. Such a map has degree one, and hence is homotopic to the identity.

Then, $h \circ f \simeq f$. But $h \circ f$ is exactly the Pontryagin–Thom collapse map associated to the framed submanifold $X = f^{-1}(q)$. Therefore

$$\mathrm{PT}(\mathrm{RP}([f])) = [f]$$

Hence PT and RP are inverse bijections. □

Let us emphasize the two directions of the correspondence:

$$\begin{aligned} (X, \varphi) \subseteq M \quad &\longmapsto \quad c_{X,\varphi} : M \to S^r \\ f : M \to S^r \quad &\longmapsto \quad f^{-1}(q) \subseteq M \end{aligned}$$

where q is a regular value of f.
In the special case $M = S^m$, the correspondence becomes

$$\Omega^{\mathrm{fr}}_{m-r}(S^m) \cong [S^m, S^r] = \pi_m(S^r)$$

Thus, homotopy groups of spheres can be interpreted geometrically as framed cobordism classes of submanifolds of spheres.
This is the fundamental bridge between differential topology and unstable homotopy theory:

$$\text{homotopy classes of maps to spheres} \quad \Longleftrightarrow \quad \text{framed cobordism classes}$$

The next section applies this viewpoint to stable stems in low dimensions, degree theory, vector fields, and the Poincaré–Hopf theorem.

7 Applications and Examples

7.1 The Freudenthal Suspension Theorem

In this subsection, we prove the sphere version of the Freudenthal suspension theorem by using the Pontryagin–Thom correspondence. The geometric idea is that the suspension homomorphism is represented by adding one trivial normal direction to a framed submanifold. The stable range then comes from the fact that, under suitable dimension bounds, a framed submanifold in $\mathbb{R}^m \oplus \mathbb{R}$ may be pushed into the hyperplane $\mathbb{R}^m$ and the extra normal direction may be split off from the framing.
For integers $n \geq 1$ and $k \geq 0$, let $\mathcal{F}^n_k$ denote the set of framed cobordism classes of compact codimension n framed submanifolds of $\mathbb{R}^{n+k}$.

Proposition 7.1. *There is a natural bijection*

$$\mathrm{PT} : \mathcal{F}^n_k \xrightarrow{\cong} \pi_{n+k}(S^n).$$

Proof. Identify S^{n+k} with the one-point compactification

$$(\mathbb{R}^{n+k})^+ = \mathbb{R}^{n+k} \sqcup \{\infty\}$$

By Theorem 6.9, framed cobordism classes of compact codimension n framed submanifolds of S^{n+k} correspond bijectively to based homotopy classes

$$[S^{n+k}, S^n]_* = \pi_{n+k}(S^n)$$

If $f : S^{n+k} \to S^n$ is based, then $f(\infty) = *$, so for any regular value $q \in S^n \setminus \{*\}$, the regular preimage $f^{-1}(q)$ is automatically disjoint from ∞. Hence, the framed submanifold associated with f lies in

$$S^{n+k} \setminus \{\infty\} \cong \mathbb{R}^{n+k}$$

Conversely, a compact framed submanifold of $\mathbb{R}^{n+k}$ determines a collapse map on $(\mathbb{R}^{n+k})^+ = S^{n+k}$ which sends ∞ to the basepoint of S^n. Therefore, the correspondence is exactly the claimed bijection. □

We next describe the suspension homomorphism in this geometric model. Write

$$\mathbb{R}^{n+k+1} = \mathbb{R}^{n+k} \oplus \mathbb{R}e$$

where e denotes the unit vector in the last coordinate direction.

Proposition 7.2. *Under the identification of Proposition 7.1, the suspension homomorphism*

$$E : \pi_{n+k}(S^n) \to \pi_{n+k+1}(S^{n+1})$$

is represented by

$$(M, \varphi) \longmapsto (M, \varphi, e)$$

where $M \subseteq \mathbb{R}^{n+k} = \mathbb{R}^{n+k} \times \{0\} \subseteq \mathbb{R}^{n+k+1}$ and (M, φ, e) means that the framing of $\nu_M(\mathbb{R}^{n+k+1})$ is obtained from the framing φ of $\nu_M(\mathbb{R}^{n+k})$ by adjoining the constant normal vector e.

Proof. Let (M, φ) be a compact codimension n framed submanifold of $\mathbb{R}^{n+k}$, and choose a framed tubular neighborhood

$$\Theta : M \times D^n \xrightarrow{\cong} U \subseteq \mathbb{R}^{n+k}$$

The Pontryagin–Thom collapse map is

$$c_{M,\varphi} : (\mathbb{R}^{n+k})^+ \to D^n/\partial D^n = S^n$$

Now regard M as a submanifold of $\mathbb{R}^{n+k+1} = \mathbb{R}^{n+k} \oplus \mathbb{R}e$. Then

$$U \times D^1 \cong M \times D^n \times D^1$$

is a framed tubular neighborhood of M in $\mathbb{R}^{n+k+1}$, and the associated framing is exactly (φ, e).
The corresponding collapse map is

$$c_{M,\varphi,e} : (\mathbb{R}^{n+k+1})^+ \to D^{n+1}/\partial D^{n+1} = S^{n+1}$$

Under the canonical identification

$$(\mathbb{R}^{n+k+1})^+ \cong \Sigma(\mathbb{R}^{n+k})^+$$

the map $c_{M,\varphi,e}$ is the suspension of $c_{M,\varphi}$. Therefore,

$$\mathrm{PT}[(M, \varphi, e)] = E\big(\mathrm{PT}[(M, \varphi)]\big)$$

which proves the claim. □

The proof of the Freudenthal theorem reduces to two geometric propositions.

Proposition 7.3 (Flattening Lemma)**.** *Let $H \cong \mathbb{R}^m$, and let $X^d \subseteq H \oplus \mathbb{R}$ be a compact smooth embedded submanifold. If $\partial X \neq \emptyset$, assume that*

$$\partial X \subseteq H \times \{0\}$$

If $m > 2d$, then there exists an ambient isotopy of $H \oplus \mathbb{R}$ carrying X into $H \times \{0\}$. If $\partial X \neq \emptyset$, the isotopy may be chosen relative to ∂X.

Proof. Write the inclusion of X as

$$j = (g, h) : X \hookrightarrow H \oplus \mathbb{R}$$

where $g : X \to H$ and $h : X \to \mathbb{R}$ are smooth maps.
For each $\lambda \in H$, define the shear diffeomorphism

$$S_\lambda : H \oplus \mathbb{R} \to H \oplus \mathbb{R}, \qquad S_\lambda(y, s) = (y + \lambda s, s)$$

Since S_λ fixes $H \times \{0\}$ pointwise, it is already relative to the boundary hyperplane. Define

$$j_\lambda := S_\lambda \circ j = (g_\lambda, h), \qquad g_\lambda(x) := g(x) + h(x)\lambda$$

We claim that for some $\lambda \in H$, the map $g_\lambda : X \to H$ is an embedding.
We first exclude the failure of immersion. Suppose that g_λ is not immersive at some $x \in X$. Then there exists $0 \neq v \in T_x X$ such that

$$(\mathrm{d}g)_x(v) + \lambda(\mathrm{d}h)_x(v) = 0$$

Since j is an embedding, we have

$$\big((\mathrm{d}g)_x(v), (\mathrm{d}h)_x(v)\big) \neq (0,0)$$

Hence $(\mathrm{d}h)_x(v) \neq 0$, and therefore

$$\lambda = -\frac{(\mathrm{d}g)_x(v)}{(\mathrm{d}h)_x(v)}$$

Let $S(TX)$ be the unit tangent sphere bundle of X, which has dimension $2d-1$. On the open set

$$U_{\mathrm{imm}} := \big\{(x,v) \in S(TX) \mid (\mathrm{d}h)_x(v) \neq 0\big\}$$

define

$$A : U_{\mathrm{imm}} \to H, \qquad A(x,v) = -\frac{(\mathrm{d}g)_x(v)}{(\mathrm{d}h)_x(v)}$$

If λ makes g_λ fail to be an immersion, then $\lambda \in \mathrm{im}(A)$. Since

$$\dim U_{\mathrm{imm}} = 2d-1 < m = \dim H$$

every point of U_{imm} is a critical point of A, and Sard's theorem implies that $\mathrm{im}(A)$ has measure zero in H.
Next, we exclude the failure of injectivity. Suppose there exist distinct points $x, y \in X$ such that $g_\lambda(x) = g_\lambda(y)$. Then

$$g(x) - g(y) + \lambda\big(h(x) - h(y)\big) = 0$$

If $h(x) = h(y)$, then also $g(x) = g(y)$, which contradicts the injectivity of j. Thus $h(x) \neq h(y)$, and

$$\lambda = -\frac{g(x) - g(y)}{h(x) - h(y)}$$

On the open subset

$$U_{\mathrm{inj}} := \big\{(x,y) \in X \times X \setminus \Delta_X \mid h(x) \neq h(y)\big\}$$

define

$$B : U_{\mathrm{inj}} \to H, \qquad B(x,y) = -\frac{g(x) - g(y)}{h(x) - h(y)}$$

If λ makes g_λ fail to be injective, then $\lambda \in \mathrm{im}(B)$. Since

$$\dim U_{\mathrm{inj}} = 2d < m = \dim H$$

every point of U_{inj} is a critical point of B, and again Sard's theorem shows that $\mathrm{im}(B)$ has measure zero.
Therefore, the bad set

$$\mathrm{im}(A) \cup \mathrm{im}(B) \subseteq H$$

has measure zero. Choose

$$\lambda \in H \setminus \big(\mathrm{im}(A) \cup \mathrm{im}(B)\big)$$

Then g_λ is an injective immersion. Since X is compact and H is Hausdorff, g_λ is an embedding.

It follows that $j_\lambda(X)$ is the graph of the smooth function

$$\widetilde{h}_0 := h \circ g_\lambda^{-1} : g_\lambda(X) \to \mathbb{R}$$

Since $g_\lambda(X) \subseteq H$ is a closed embedded submanifold, $\widetilde{h}_0$ extends to a smooth function

$$\widetilde{h} : H \to \mathbb{R}$$

If $\partial X \neq \emptyset$, then $\widetilde{h}_0 = 0$ on $g_\lambda(\partial X)$ because $\partial X \subseteq H \times \{0\}$, and we may choose the extension so that

$$\widetilde{h}|_{g_\lambda(\partial X)} = 0$$

Now define

$$T_t : H \oplus \mathbb{R} \to H \oplus \mathbb{R}, \qquad T_t(y,s) = \big(y, s - t\widetilde{h}(y)\big), \qquad 0 \leq t \leq 1$$

Each T_t is a diffeomorphism, with inverse

$$T_t^{-1}(y,s) = \big(y, s + t\widetilde{h}(y)\big)$$

At $t = 1$, the graph of $\widetilde{h}_0$ is sent to $H \times \{0\}$. Since the path $t \mapsto S_{t\lambda}$ is an ambient isotopy from the identity to S_λ, the composition of these two isotopies carries X into $H \times \{0\}$. If $\partial X \neq \emptyset$, then both isotopies fix ∂X: the shear fixes the whole hyperplane $H \times \{0\}$, and T_t fixes $g_\lambda(\partial X) \times \{0\}$ because $\widetilde{h} = 0$ there. □

Proposition 7.4 (Splitting Lemma)**.** *Let $X^d \subseteq H \cong \mathbb{R}^{d+c}$ be a compact smooth submanifold, and regard X as a submanifold of $H \oplus \mathbb{R}e$. Let*

$$\Phi = (v_1, \ldots, v_{c+1})$$

be an orthonormal framing of the normal bundle $\nu_X(H \oplus \mathbb{R}e)$. If $d < c$, then Φ is homotopic through orthonormal framings to a framing of the form $(\psi_1, \ldots, \psi_c, e)$, where $(\psi_1, \ldots, \psi_c)$ is an orthonormal framing of $\nu_X(H)$.
If $\partial X \neq \emptyset$ and $v_{c+1} = e$ on ∂X, then the homotopy may be chosen relative to ∂X.

Proof. Since Φ is an orthonormal frame of the normal bundle, the ambient vector e may be written uniquely as

$$e = \sum_{i=1}^{c+1} u_i(x) v_i(x), \qquad x \in X$$

where $u = (u_1, \ldots, u_{c+1}) : X \to S^c$ is a smooth map. If $\partial X \neq \emptyset$ and $v_{c+1} = e$ on ∂X, then

$$u|_{\partial X} = e_{c+1} \in S^c$$

Since X is a smooth d-manifold, it has the homotopy type of a CW complex of dimension d. Because $d < c$ and S^c is $(c-1)$-connected, every map $X \to S^c$ is null-homotopic. Hence u is homotopic to the constant map e_{c+1}. In the relative case, the same argument applied to the quotient $X/\partial X$ shows that u is homotopic to the constant map e_{c+1} relative to ∂X.
Now consider the Stiefel fibration

$$O(c) \longrightarrow O(c+1) \xrightarrow{p} S^c, \qquad p(A) = Ae_{c+1}$$

Choose a homotopy

$$u_t : X \to S^c, \qquad 0 \leq t \leq 1$$

such that

$$u_0 = e_{c+1}, \qquad u_1 = u$$

and in the relative case

$$u_t|_{\partial X} = e_{c+1} \qquad \forall t \in [0,1]$$

Since p is a fiber bundle, it has the homotopy lifting property. Therefore, there exists a lift $A_t : X \to O(c+1)$ such that

$$A_0 = I, \qquad A_t e_{c+1} = u_t$$

In the relative case, the lift may be chosen so that

$$A_t|_{\partial X} = I \qquad \forall t \in [0,1]$$

Define a new homotopy of orthonormal framings by

$$\Phi_t := \Phi \cdot A_t^{-1}$$

Here, the matrix A_t^{-1} acts on the ordered frame Φ on the right. At $t = 1$, the coordinates of the ambient vector e with respect to the framing Φ_1 are $A_1^{-1}u = e_{c+1}$, because $u = A_1 e_{c+1}$. Therefore, the last vector of Φ_1 is exactly e.
Since Φ_1 is orthonormal and its last vector is e, its first c vectors are orthogonal to e, hence they lie in the hyperplane $H = e^{\perp}$. They are also normal to TX, so they form an orthonormal framing of

$$\nu_X(H) = \nu_X(H \oplus \mathbb{R}e) \cap H$$

Thus, $\Phi_1 = (\psi_1, \ldots, \psi_c, e)$ for some orthonormal framing $(\psi_1, \ldots, \psi_c)$ of $\nu_X(H)$, as required. □

We may now prove the Freudenthal theorem.

Theorem 7.5 (Freudenthal Suspension Theorem for Spheres)**.** *For $n \geq 1$, the suspension homomorphism*

$$E : \pi_q(S^n) \to \pi_{q+1}(S^{n+1})$$

is an isomorphism for $q \leq 2n-2$ and a surjection for $q = 2n-1$.

Proof. Write $q = n + k$. Then

$$q \leq 2n-1 \iff k < n, \qquad q \leq 2n-2 \iff k+1 < n.$$

Thus it is enough to prove that

$$E : \pi_{n+k}(S^n) \to \pi_{n+k+1}(S^{n+1})$$

is surjective when $k < n$ and injective when $k+1 < n$.
We first prove surjectivity. Let

$$\alpha \in \pi_{n+k+1}(S^{n+1}).$$

By Proposition 7.1, α is represented by a compact codimension $(n+1)$ framed k-submanifold

$$(M, \Phi) \subseteq \mathbb{R}^{n+k+1} = \mathbb{R}^{n+k} \oplus \mathbb{R}e.$$

After applying fiberwise Gram–Schmidt, we may assume that Φ is orthonormal. Since

$$n + k > 2k$$

by the assumption $k < n$, Proposition 7.3 applies with $d = k$ and $m = n + k$. Hence, after an ambient isotopy of $\mathbb{R}^{n+k+1}$, we may assume that $M \subseteq \mathbb{R}^{n+k} \times \{0\}$. Transport the framing Φ along this isotopy.
Now apply Proposition 7.4 with $d = k$ and $c = n$. Since $k < n$, the framing Φ is homotopic to one of the form (φ, e), where φ is a framing of the codimension n normal bundle of M in $\mathbb{R}^{n+k}$.
Therefore, (M, φ) determines a class $\beta \in \pi_{n+k}(S^n)$. By Proposition 7.2, the suspended class $E(\beta)$ is represented by (M, φ, e), which is framed cobordant to (M, Φ). Hence, $E(\beta) = \alpha$. This proves surjectivity when $k < n$.

We now prove injectivity. Suppose $\alpha_0, \alpha_1 \in \pi_{n+k}(S^n)$ satisfy

$$E(\alpha_0) = E(\alpha_1)$$

Choose compact codimension n framed k-submanifolds

$$(M_0, \varphi_0), \qquad (M_1, \varphi_1) \subseteq \mathbb{R}^{n+k}$$

representing α_0 and α_1, and after fiberwise Gram–Schmidt assume that φ_0 and φ_1 are orthonormal. Then $E(\alpha_0)$ and $E(\alpha_1)$ are represented by (M_0, φ_0, e) and (M_1, φ_1, e) inside $\mathbb{R}^{n+k+1} = \mathbb{R}^{n+k} \oplus \mathbb{R}e$.
Since these two classes are equal, there exists a framed cobordism

$$(W, \Phi) \subseteq \left(\mathbb{R}^{n+k+1}\right) \times [0,1]$$

from (M_0, φ_0, e) to (M_1, φ_1, e). By orthonormalizing Φ fiberwise and keeping the boundary framing fixed, we may assume that Φ is orthonormal and that on the boundary the last normal vector is e.
Rewrite the ambient space as

$$\left(\mathbb{R}^{n+k} \times [0,1]\right) \oplus \mathbb{R}e$$

Since $\dim W = k+1$ and $\dim\left(\mathbb{R}^{n+k} \times [0,1]\right) = n+k+1$, the inequality $k+1 < n$ implies

$$n + k + 1 > 2(k+1)$$

Hence Proposition 7.3 applies, relative to the boundary, and carries W into

$$\mathbb{R}^{n+k} \times [0,1]$$

Transport the framing along this isotopy.
Now apply the relative form of Proposition 7.4 with $d = k+1$ and $c = n$. Since $k+1 < n$, the framing Φ is homotopic relative to the boundary to a framing of the form (Ψ, e), where Ψ is a framing of the codimension n normal bundle of W in $\mathbb{R}^{n+k} \times [0,1]$.
Therefore, (W, Ψ) is a framed cobordism in $\mathbb{R}^{n+k} \times [0,1]$ from (M_0, φ_0) to (M_1, φ_1). By Proposition 7.1, this means $\alpha_0 = \alpha_1$. Hence E is injective when $k+1 < n$.
Combining the two parts, E is an isomorphism for $q \leq 2n-2$ and a surjection for $q = 2n-1$. □

A standard consequence is the stability of the homotopy groups $\pi_{n+k}(S^n)$ once n is sufficiently large relative to k.

Corollary 7.6. *Fix $k \geq 0$. Then the suspension map*

$$E : \pi_{n+k}(S^n) \to \pi_{n+k+1}(S^{n+1})$$

is an isomorphism for every $n > k+1$. Consequently, the groups $\pi_{n+k}(S^n)$ are independent of n in this range.

Proof. If $n > k+1$, then $k+1 < n$, so Theorem 7.5 implies that the suspension map is an isomorphism. Repeating suspension, therefore, identifies all the groups in the stable range. □

7.2 Calculation of $\pi_{n+1}(S^n)$

We first calculate $\pi_{n+1}(S^n)$. For $n \geq 3$, the answer is $\mathbb{Z}/2$; the low-dimensional cases $n = 1$ and $n = 2$ are recorded below.
The Pontryagin–Thom construction gives the geometric interpretation

$$\pi_{n+1}(S^n) = [S^{n+1}, S^n]$$

If $q \in S^n$ is a regular value of a smooth map $f : S^{n+1} \to S^n$, then $f^{-1}(q)$ is a closed 1-dimensional framed submanifold of S^{n+1}.
Thus, for $n \geq 3$, the problem is equivalent to classifying closed framed 1-manifolds up to framed cobordism. The codimension is at least 3, so the framed cobordism group is already stable. We denote it by Ω_1^{fr}.

Theorem 7.7. *The framed bordism group of closed* 1*-manifolds is* $\Omega_1^{\mathrm{fr}} \cong \mathbb{Z}/2$.

Proof. Let (L, φ) be a closed framed 1-manifold. Since every closed 1-manifold is a disjoint union of circles, we may write

$$L = S_1^1 \sqcup \cdots \sqcup S_k^1$$

A normal framing of L determines a stable framing of TL. Indeed, for a framed submanifold $L \subseteq S^{n+1}$, the stable triviality

$$TS^{n+1} \oplus \mathbb{R} \cong \mathbb{R}^{n+2}$$

and the normal framing $\nu_L(S^{n+1}) \cong L \times \mathbb{R}^n$ imply

$$TL \oplus \mathbb{R}^n \oplus \mathbb{R} \cong L \times \mathbb{R}^{n+2}$$

Hence, each component S_i^1 inherits a stable framing.

There are exactly two stable framings of S^1 up to homotopy. One extends over a disk D^2, and the other does not. Define

$$\epsilon(S_i^1, \varphi_i) = \begin{cases} 0, & \text{if the induced stable framing extends over } D^2, \\ 1, & \text{otherwise} \end{cases}$$

Then define

$$\alpha(L, \varphi) := \sum_{i=1}^{k} \epsilon(S_i^1, \varphi_i) \in \mathbb{Z}/2$$

This number is invariant under framed cobordism. Indeed, if (L_0, φ_0) and (L_1, φ_1) are framed cobordant by a framed surface W, then the framing of W gives a stable framing of TW. Therefore, the boundary stable framings on L_0 and L_1 differ by a stable framed cobordism. The obstruction to extending the boundary stable framing over W has zero total boundary contribution, and hence

$$\alpha(L_0, \varphi_0) = \alpha(L_1, \varphi_1)$$

The invariant α is surjective because a circle with the non-bounding stable framing has $\alpha = 1$. It remains to show that α is complete.

If $\alpha(L, \varphi) = 0$, then the number of components with non-bounding stable framing is even. A component with bounding framing bounds a framed disk. Two components with non-bounding framings together bound a framed annulus. Therefore, all components of L can be removed by framed cobordisms, and (L, φ) is framed null-cobordant.

Thus, the framed cobordism class is completely determined by $\alpha \in \mathbb{Z}/2$. Hence, $\Omega_1^{\mathrm{fr}} \cong \mathbb{Z}/2$. □

Under Pontryagin–Thom, the nonzero element of Ω_1^{fr} corresponds to the suspension of the Hopf map. Let $\eta_2 : S^3 \to S^2$ be the Hopf fibration. For $n \geq 3$, define

$$\eta_n := \Sigma^{n-2}\eta_2 \in \pi_{n+1}(S^n)$$

This element is represented by a framed circle with the non-bounding stable framing.

Theorem 7.8. *For* $n \geq 3$, $\pi_{n+1}(S^n) \cong \mathbb{Z}/2$. *The nonzero element is represented by* η_n, *and* $2\eta_n = 0$.

Proof. By the Pontryagin–Thom construction, a class in $\pi_{n+1}(S^n)$ is represented by a closed codimension n framed 1-submanifold of S^{n+1}. Since $n \geq 3$, this framed cobordism group is the stable framed bordism group Ω_1^{fr}.

By Theorem 7.7, $\Omega_1^{\mathrm{fr}} \cong \mathbb{Z}/2$. Therefore

$$\pi_{n+1}(S^n) \cong \mathbb{Z}/2$$

The generator is the class represented by a framed circle whose induced stable framing is non-bounding. Under Pontryagin–Thom, this is exactly the suspended Hopf class η_n. Since the group is $\mathbb{Z}/2$, we have $2\eta_n = 0$. □

For completeness, the low-dimensional cases are

$$\pi_2(S^1) = 0, \qquad \pi_3(S^2) \cong \mathbb{Z}$$

The second group is generated by the Hopf fibration $\eta_2 : S^3 \to S^2$. Thus the stable answer $\mathbb{Z}/2$ begins at $n = 3$.

Therefore,

$$\pi_{n+1}(S^n) \cong \begin{cases} 0, & n = 1, \\ \mathbb{Z}, & n = 2, \\ \mathbb{Z}/2, & n \geq 3 \end{cases}$$

7.3 Calculation of $\pi_{n+2}(S^n)$

We now calculate $\pi_{n+2}(S^n)$. For $n \geq 4$, the Pontryagin–Thom construction identifies this group with the framed bordism group of closed framed surfaces:

$$\pi_{n+2}(S^n) \cong \Omega_2^{\text{fr}}$$

The answer is again $\mathbb{Z}/2$, but the invariant is subtler than in dimension 1. It is the Arf invariant of the spin structure determined by the framing.

Let (Σ, φ) be a closed framed surface. The normal framing and the stable triviality of the sphere determine a stable framing of $T\Sigma$, and hence a spin structure on Σ. A spin structure on an oriented closed surface determines a quadratic refinement $q_\varphi : H_1(\Sigma; \mathbb{F}_2) \to \mathbb{F}_2$ of the mod-2 intersection pairing. This means

$$q_\varphi(a + b) = q_\varphi(a) + q_\varphi(b) + a \cdot b$$

for all $a, b \in H_1(\Sigma; \mathbb{F}_2)$.

Choose a symplectic basis $a_1, b_1, \ldots, a_g, b_g$ of $H_1(\Sigma; \mathbb{F}_2)$, so that $a_i \cdot b_i = 1$ and all other intersections vanish. The Arf invariant of q_φ is

$$\text{Arf}(q_\varphi) = \sum_{i=1}^{g} q_\varphi(a_i) q_\varphi(b_i) \in \mathbb{F}_2$$

This number is independent of the chosen symplectic basis.

Theorem 7.9. *The framed bordism group of closed surfaces is*

$$\Omega_2^{\text{fr}} \cong \mathbb{Z}/2$$

The isomorphism is given by the Arf invariant.

Proof. We define $\text{Arf}(\Sigma, \varphi) := \text{Arf}(q_\varphi) \in \mathbb{F}_2$. First, this is invariant under framed cobordism. If (Σ_0, φ_0) and (Σ_1, φ_1) are framed cobordant, then there exists a framed 3-manifold W with

$$\partial W = \Sigma_0 \sqcup \Sigma_1$$

The framing of W induces a spin cobordism between the spin structures on Σ_0 and Σ_1. The Arf invariant is invariant under spin cobordism of surfaces. Hence

$$\text{Arf}(\Sigma_0, \varphi_0) = \text{Arf}(\Sigma_1, \varphi_1)$$

We next show that the Arf invariant is complete. Let (Σ, φ) be a closed framed surface. If there exists a simple closed curve $\gamma \subseteq \Sigma$ with $q_\varphi([\gamma]) = 0$, then the framing along γ is the bounding one. Hence, one can perform framed surgery along γ. This surgery removes a handle and preserves the framed bordism class.

By repeating framed surgeries, we can reduce Σ to a standard form determined only by the Arf invariant. If $\mathrm{Arf}(q_\varphi) = 0$, then one can choose enough curves with $q_\varphi = 0$ to kill all handles. The result is a framed 2-sphere. Since $\pi_2(SO) = 0$, the stable framing on this 2-sphere extends over a 3-ball. Hence, the original framed surface is framed null-cobordant.

If $\mathrm{Arf}(q_\varphi) = 1$, then framed surgeries reduce (Σ, φ) to a framed torus (T^2, φ_T) with a symplectic basis a, b satisfying $q_{\varphi_T}(a) = q_{\varphi_T}(b) = 1$. This torus has Arf invariant 1, and hence it is not framed null-cobordant.

Therefore, every framed surface is framed cobordant either to the empty surface or to this framed torus. Since the Arf invariant is additive under disjoint union, two copies of the framed torus have Arf invariant 0 and are framed null-cobordant. Thus, the framed bordism group is $\mathbb{Z}/2$. □

The nonzero element of Ω_2^{fr} corresponds under Pontryagin–Thom to the stable class usually denoted by η^2. Here η is the stable Hopf element, and η^2 is its product in the stable homotopy ring.

Theorem 7.10. *For $n \geq 4$,*

$$\pi_{n+2}(S^n) \cong \mathbb{Z}/2$$

The nonzero element is represented by η_n^2.

Proof. Let $f : S^{n+2} \to S^n$ be a smooth representative of a homotopy class. For a regular value $q \in S^n$, the preimage $\Sigma = f^{-1}(q)$ is a closed 2-dimensional framed submanifold of S^{n+2}. Since $n \geq 4$, the codimension is large enough for the framed cobordism group to be stable. Thus Pontryagin–Thom gives $\pi_{n+2}(S^n) \cong \Omega_2^{\mathrm{fr}}$. By Theorem 7.9, $\Omega_2^{\mathrm{fr}} \cong \mathbb{Z}/2$. Hence

$$\pi_{n+2}(S^n) \cong \mathbb{Z}/2.$$

The generator is represented by a framed torus with Arf invariant 1. Under Pontryagin–Thom, this generator is the element η_n^2. □

It remains to record the two low-dimensional cases $n = 2$ and $n = 3$. They have the same answer, although they do not lie in the stable range used above.

First, consider the Hopf fibration $S^1 \hookrightarrow S^3 \twoheadrightarrow S^2$. The long exact sequence in homotopy gives an isomorphism

$$\pi_4(S^3) \cong \pi_4(S^2)$$

because $\pi_4(S^1) = 0$ and $\pi_3(S^1) = 0$. By Theorem 7.8, applied to S^3, we have $\pi_4(S^3) \cong \mathbb{Z}/2$. Therefore $\pi_4(S^2) \cong \mathbb{Z}/2$.

Next, consider the quaternionic Hopf fibration $S^3 \hookrightarrow S^7 \twoheadrightarrow S^4$. The long exact sequence gives

$$\cdots \longrightarrow \pi_6(S^7) \longrightarrow \pi_6(S^4) \longrightarrow \pi_5(S^3) \longrightarrow \pi_5(S^7) \longrightarrow \cdots$$

Since $\pi_6(S^7) = 0$ and $\pi_5(S^7) = 0$, we get an isomorphism

$$\pi_6(S^4) \cong \pi_5(S^3)$$

By Theorem 7.10, applied to $n = 4$, we have $\pi_6(S^4) \cong \mathbb{Z}/2$. Hence $\pi_5(S^3) \cong \mathbb{Z}/2$.

Therefore, for every $n \geq 2$,

$$\pi_{n+2}(S^n) \cong \mathbb{Z}/2$$

Geometrically, this means that a class in $\pi_{n+2}(S^n)$ is detected by a framed surface, and the only obstruction to being framed null-cobordant is the Arf invariant. Thus, the two possible classes are represented by

$$\mathrm{Arf} = 0 \qquad \text{and} \qquad \mathrm{Arf} = 1.$$

The class with $\mathrm{Arf} = 1$ is the element η^2.

7.4 J-Homomorphism

This subsection will be completed in a future revision.

7.5 Degree, Vector Fields, and the Poincaré–Hopf Theorem

The purpose of this subsection is to explain the Poincaré–Hopf theorem from the viewpoint of degree theory, zero loci of sections, and cobordism. The basic philosophy is that the degree of a map counts regular preimages as oriented zero-dimensional manifolds, while the index of a vector field counts zeros of a section of the tangent bundle by local degree.

Throughout this subsection, unless otherwise stated, M^n denotes a closed oriented smooth manifold.

Definition 7.11 (Degree via Regular Values)**.** Let M^n and N^n be closed oriented smooth manifolds, and suppose that N is connected. Let $f : M \to N$ be a smooth map. If $y \in N$ is a regular value of f, then $f^{-1}(y)$ is a finite set. For each $x \in f^{-1}(y)$, the differential

$$(\mathrm{d}f)_x : T_xM \to T_yN$$

is a linear isomorphism. Define

$$\epsilon_x(f) = \begin{cases} +1, & (\mathrm{d}f)_x \text{ preserves orientation,} \\ -1, & (\mathrm{d}f)_x \text{ reverses orientation} \end{cases}$$

The degree of f is defined by

$$\deg(f) := \sum_{x \in f^{-1}(y)} \epsilon_x(f)$$

Proposition 7.12 (Degree as a Zero-Dimensional Cobordism Invariant)**.** *Let $f_0, f_1 : M^n \to N^n$ be smooth maps between closed oriented smooth manifolds, and suppose that f_0 and f_1 are smoothly homotopic. Then* $\deg(f_0) = \deg(f_1)$.

Proof. Let $F : M \times [0,1] \to N$ be a smooth homotopy from f_0 to f_1, i.e.

$$F(x,0) = f_0(x), \qquad F(x,1) = f_1(x)$$

By a small perturbation of F relative to the boundary if necessary, we may assume that a chosen regular value $y \in N$ is regular for F, f_0, and f_1.

Consider $W := F^{-1}(y)$. Since y is a regular value of F, the preimage theorem gives that W is a compact oriented 1-dimensional manifold. Moreover, its boundary is

$$\partial W = f_1^{-1}(y) - f_0^{-1}(y)$$

where the minus sign comes from the boundary orientation on $M \times [0,1]$.

A compact oriented 1-manifold is a finite disjoint union of circles and compact intervals. Every compact interval has two boundary points with opposite signs. Hence, the algebraic boundary count of W is zero. Therefore

$$\sum_{x \in f_1^{-1}(y)} \epsilon_x(f_1) - \sum_{x \in f_0^{-1}(y)} \epsilon_x(f_0) = 0$$

Thus, $\deg(f_1) = \deg(f_0)$. □

Remark. This proof shows that the degree is fundamentally a cobordism invariant. The regular preimage $f^{-1}(y)$ is an oriented zero-dimensional manifold, and the homotopy F produces an oriented cobordism between the two zero-dimensional manifolds

$$f_0^{-1}(y) \qquad \text{and} \qquad f_1^{-1}(y)$$

Since $\Omega_0^{SO} \cong \mathbb{Z}$, the cobordism class of an oriented finite set is exactly its signed count.

Definition 7.13 (Local Degree)**.** Let $U \subseteq \mathbb{R}^n$ be open, and let $g : U \to \mathbb{R}^n$ be a smooth map. Suppose that $a \in U$ is an isolated zero of g, i.e. $g(a) = 0$, $g(x) \neq 0$ for $0 < \|x - a\| \ll 1$. Choose $\varepsilon > 0$ sufficiently small such that $\overline{B_\varepsilon(a)} \subseteq U$ and $g(x) \neq 0$ for $x \in \partial B_\varepsilon(a)$. The local degree of g at a is defined by

$$\deg_a(g) := \deg\left(\frac{g}{\|g\|} : \partial B_\varepsilon(a) \to S^{n-1}\right)$$

Proposition 7.14 (Local Degree at a Nondegenerate Zero)**.** *Let $g : U \subseteq \mathbb{R}^n \to \mathbb{R}^n$ be smooth, and suppose that $a \in U$ is a nondegenerate zero, i.e.*

$$g(a) = 0, \qquad Dg_a : \mathbb{R}^n \to \mathbb{R}^n \text{ is an isomorphism}$$

Then, $\deg_a(g) = \operatorname{sgn}\det(Dg_a)$.

Proof. Since Dg_a is invertible, the inverse function theorem implies that a is an isolated zero. For sufficiently small $\varepsilon > 0$, the map g on $\partial B_\varepsilon(a)$ is homotopic through nonvanishing maps to its linear approximation $x \mapsto Dg_a(x - a)$. Hence,

$$\deg_a(g) = \deg\left(\frac{Dg_a(x-a)}{\|Dg_a(x-a)\|} : \partial B_\varepsilon(a) \to S^{n-1}\right)$$

The last degree is $+1$ if Dg_a preserves orientation and -1 if Dg_a reverses orientation. Therefore, $\deg_a(g) = \operatorname{sgn}\det(Dg_a)$, which proves the proposition. □

Definition 7.15 (Index of a Vector Field)**.** Let $X \in \Gamma(TM)$ be a smooth vector field on an oriented smooth manifold M^n, and let $p \in M$ be an isolated zero of X. Choose an oriented coordinate chart

$$(U, \varphi) = (U, x^1, \ldots, x^n)$$

around p such that $\varphi(p) = 0$. Write

$$X|_U = \sum_{i=1}^{n} X^i \frac{\partial}{\partial x^i}$$

This gives a smooth map

$$X_\varphi : \varphi(U) \to \mathbb{R}^n, \qquad X_\varphi = (X^1, \ldots, X^n)$$

The index of X at p is defined by $\operatorname{ind}_p(X) := \deg_0(X_\varphi)$.

Proposition 7.16 (Well-Definedness and Nondegenerate Formula for the Index)**.** *The integer* $\operatorname{ind}_p(X)$ *is independent of the chosen oriented coordinate chart. Moreover, if p is a nondegenerate zero of X, then*

$$\operatorname{ind}_p(X) = \operatorname{sgn}\det(\nabla X)_p,$$

where ∇ is any connection on TM.

Proof. First consider the independence of coordinates. If two oriented charts are chosen, then the corresponding local maps differ by composition with orientation-preserving local diffeomorphisms on the source and target. Since the degree is invariant under orientation-preserving changes of coordinates, the local degree is unchanged. Hence $\operatorname{ind}_p(X)$ is well-defined.

Now suppose that p is a nondegenerate zero. Since $X(p) = 0$, the linear map

$$(\nabla X)_p : T_pM \to T_pM$$

is independent of the choice of connection. In a local coordinate trivialization, this linear map is represented by the Jacobian matrix of the local map X_φ at $\varphi(p) = 0$. Therefore, by the previous proposition, $\operatorname{ind}_p(X) = \operatorname{sgn}\det(\nabla X)_p$. □

Proposition 7.17 (Zero Locus of a Transverse Section)**.** *Let $E \to M^m$ be a smooth rank r vector bundle, and let $s \in \Gamma(E)$ be a smooth section transverse to the zero section. Then $Z(s) := s^{-1}(0)$ is a smooth embedded submanifold of M of dimension*

$$\dim Z(s) = m - r$$

Moreover, for every $p \in Z(s)$, there is a short exact sequence

$$0 \longrightarrow T_pZ(s) \hookrightarrow T_pM \xrightarrow{D_ps} E_p \longrightarrow 0$$

and hence a canonical normal bundle isomorphism $\nu_{Z(s)}(M) \cong E|_{Z(s)}$.

Proof. Locally trivialize E over an open set $U \subseteq M$: $E|_U \cong U \times \mathbb{R}^r$. Under this trivialization, the section s is represented by a smooth map $s_U : U \to \mathbb{R}^r$. The transversality of s to the zero section is exactly the statement that $0 \in \mathbb{R}^r$ is a regular value of s_U. Therefore, by the preimage theorem (Theorem 4.2), $Z(s) \cap U = s_U^{-1}(0)$ is a smooth submanifold of U of dimension $m - r$.

At $p \in Z(s)$, the tangent space is

$$T_pZ(s) = \ker(D_ps)$$

where $D_ps : T_pM \to E_p$ is the vertical derivative of the section at the zero point. Since s is transverse to the zero section, D_ps is surjective. Hence, we obtain the short exact sequence

$$0 \longrightarrow T_pZ(s) \hookrightarrow T_pM \xrightarrow{D_ps} E_p \longrightarrow 0$$

Taking the quotient by $T_pZ(s)$ gives $T_pM/T_pZ(s) \cong E_p$. Therefore, $\nu_{Z(s)}(M) \cong E|_{Z(s)}$. □

Remark. If E is equipped with a framing $E \cong M \times \mathbb{R}^r$, then the canonical isomorphism

$$\nu_{Z(s)}(M) \cong E|_{Z(s)}$$

gives a framing of $Z(s) \subseteq M$. This is the basic geometric mechanism behind the Pontryagin–Thom construction. For the Poincaré–Hopf theorem, however, the relevant bundle is $E = TM$, which is not necessarily trivial. Therefore the correct object is not necessarily a framed zero locus, but a zero locus carrying the normal data coming from TM.

Definition 7.18 (Algebraic Zero Count)**.** Let $E \to M^n$ be an oriented rank n vector bundle over a closed oriented smooth manifold M^n. Let $s \in \Gamma(E)$ be transverse to the zero section. Then $Z(s)$ is a finite set.

For $p \in Z(s)$, define

$$\epsilon_p(s) = \begin{cases} +1, & D_ps : T_pM \to E_p \text{ preserves orientation,} \\ -1, & D_ps : T_pM \to E_p \text{ reverses orientation.} \end{cases}$$

The algebraic zero count of s is

$$\#^{\text{alg}} Z(s) := \sum_{p \in Z(s)} \epsilon_p(s)$$

Proposition 7.19 (Cobordism Invariance of the Algebraic Zero Count)**.** *Let $E \to M^n$ be an oriented rank n vector bundle over a closed oriented smooth manifold. If $s_0, s_1 \in \Gamma(E)$ are transverse to the zero section, then*

$$\#^{\text{alg}} Z(s_0) = \#^{\text{alg}} Z(s_1)$$

Proof. Choose a smooth homotopy of sections $s_t \in \Gamma(E)$ with $t \in [0,1]$, from s_0 to s_1. Equivalently, define a section $S \in \Gamma(\pi_M^* E)$ over $M \times [0,1]$ by $S(x,t) = s_t(x)$. By a small perturbation relative to the boundary, we may assume that S is transverse to the zero section of $\pi_M^* E$.

Then, $W := Z(S) = S^{-1}(0)$ is a compact oriented 1-dimensional manifold. Its boundary is $\partial W = Z(s_1) - Z(s_0)$. Since every compact oriented 1-manifold has algebraic boundary count zero, we have

$$\#^{\text{alg}} Z(s_1) - \#^{\text{alg}} Z(s_0) = 0$$

Therefore, $\#^{\text{alg}} Z(s_0) = \#^{\text{alg}} Z(s_1)$. □

Definition 7.20 (Euler Number via Zero Loci)**.** Let $E \to M^n$ be an oriented rank n vector bundle over a closed oriented smooth manifold M^n. The Euler number of E is defined by

$$\langle e(E), [M] \rangle := \#^{\text{alg}} Z(s),$$

where $s \in \Gamma(E)$ is any section transverse to the zero section.

Remark. The previous proposition proves that the definition is independent of the chosen transverse section. In cohomological language, the zero locus of a transverse section represents the Poincaré dual of the Euler class:

$$[Z(s)] = \text{PD}(e(E))$$

When $\text{rank}\, E = \dim M$, this becomes the numerical identity

$$\#^{\text{alg}} Z(s) = \langle e(E), [M] \rangle$$

Proposition 7.21 (Vector Fields as Sections of the Tangent Bundle)**.** *Let $X \in \Gamma(TM)$ be a vector field on a closed oriented smooth manifold M^n. Suppose that X is transverse to the zero section. Then $Z(X)$ is finite, all zeros are nondegenerate, and*

$$\sum_{p \in Z(X)} \text{ind}_p(X) = \langle e(TM), [M] \rangle$$

Proof. A vector field is precisely a section of the tangent bundle $X \in \Gamma(TM)$. Since X is transverse to the zero section, the zero locus $Z(X) = X^{-1}(0)$ is a zero-dimensional submanifold of M, hence a finite set.

At every zero $p \in Z(X)$, the vertical derivative $D_pX : T_pM \to T_pM$ is an isomorphism. Hence p is a nondegenerate zero. By the nondegenerate index formula, $\text{ind}_p(X) = \text{sgn}\det(D_pX)$. This is exactly the sign $\epsilon_p(X)$ appearing in the algebraic zero count of the section X. Therefore

$$\sum_{p \in Z(X)} \text{ind}_p(X) = \#^{\text{alg}} Z(X)$$

By the definition of the Euler number, $\#^{\text{alg}} Z(X) = \langle e(TM), [M] \rangle$. Thus

$$\sum_{p \in Z(X)} \text{ind}_p(X) = \langle e(TM), [M] \rangle$$

which completes the proof. □

Proposition 7.22 (Additivity of the Local Index)**.** *Let X be a vector field on M, and let $p \in M$ be an isolated zero. Choose a coordinate ball B around p such that X has no zero on ∂B. Suppose that X' is another vector field on B such that $X' = X$ near ∂B and suppose that X' has only nondegenerate zeros in B. Then*

$$\text{ind}_p(X) = \sum_{q \in Z(X') \cap B} \text{ind}_q(X')$$

Proof. In a local trivialization over B, the vector fields X and X' are represented by maps $B \to \mathbb{R}^n$. Since $X' = X$ near ∂B, the normalized maps $X / \|X\|$ and $X' / \|X'\|$ agree on ∂B. Therefore, their degrees on ∂B are the same.

Let $q_1, \ldots, q_k$ be the zeros of X' in B. Choose pairwise disjoint small balls B_i around q_i. On the compact manifold with boundary

$$B \setminus \bigcup_{i=1}^{k} \text{int}(B_i),$$

the vector field X' is nowhere zero. Hence the normalized map

$$\frac{X'}{\|X'\|} : B \setminus \bigcup_{i=1}^{k} \text{int}(B_i) \to S^{n-1}$$

gives a cobordism between the outer boundary map on ∂B and the inner boundary maps on the ∂B_i. Therefore the degree on the outer boundary equals the sum of the degrees on the inner boundaries:

$$\deg\left(\frac{X'}{\|X'\|} : \partial B \to S^{n-1}\right) = \sum_{i=1}^{k} \deg\left(\frac{X'}{\|X'\|} : \partial B_i \to S^{n-1}\right)$$

The left-hand side is $\mathrm{ind}_p(X)$, while the i-th term on the right-hand side is $\mathrm{ind}_{q_i}(X')$. Hence

$$\mathrm{ind}_p(X) = \sum_{q \in Z(X') \cap B} \mathrm{ind}_q(X')$$

which completes the proof. □

Theorem 7.23 (Poincaré–Hopf Theorem)**.** *Let M^n be a closed oriented smooth manifold, and let $X \in \Gamma(TM)$ be a smooth vector field with isolated zeros. Then*

$$\sum_{p \in Z(X)} \mathrm{ind}_p(X) = \chi(M).$$

Proof. We first prove the theorem for vector fields transverse to the zero section. If $X \pitchfork 0_M$, then by the previous proposition,

$$\sum_{p \in Z(X)} \mathrm{ind}_p(X) = \langle e(TM), [M] \rangle.$$

Thus, it remains to compute the Euler number of the tangent bundle.

Choose a Morse function $f : M \to \mathbb{R}$ and a Riemannian metric g on M. Let $Y = \nabla f$ be the gradient vector field. The zeros of Y are precisely the critical points of f: $Z(Y) = \mathrm{Crit}(f)$. Since f is Morse, every critical point is nondegenerate, so Y is transverse to the zero section.

Let $p \in \mathrm{Crit}(f)$, and let $\lambda(p)$ be the Morse index of p. At p, the linearization of ∇f is given by

$$D_p(\nabla f) = g_p^{-1} \circ \mathrm{Hess}_p(f).$$

Since g_p^{-1} is positive definite, it has a positive determinant. Therefore, the sign of $\det D_p(\nabla f)$ is the same as the sign of the determinant of the Hessian of f at p.

By the Morse lemma, the Hessian of f at p has $\lambda(p)$ negative eigenvalues and $n - \lambda(p)$ positive eigenvalues. Hence, $\mathrm{ind}_p(\nabla f) = (-1)^{\lambda(p)}$. Therefore,

$$\sum_{p \in Z(\nabla f)} \mathrm{ind}_p(\nabla f) = \sum_{p \in \mathrm{Crit}(f)} (-1)^{\lambda(p)}.$$

Let c_k be the number of critical points of f of Morse index k. Then

$$\sum_{p \in \mathrm{Crit}(f)} (-1)^{\lambda(p)} = \sum_{k=0}^{n} (-1)^k c_k.$$

By Morse theory, M admits a CW decomposition with exactly c_k cells of dimension k. Hence

$$\chi(M) = \sum_{k=0}^{n} (-1)^k c_k.$$

Consequently,

$$\langle e(TM), [M] \rangle = \sum_{p \in Z(\nabla f)} \mathrm{ind}_p(\nabla f) = \chi(M).$$

Thus, for every vector field transverse to the zero section,

$$\sum_{p \in Z(X)} \operatorname{ind}_p(X) = \chi(M).$$

Now let X be any vector field with isolated zeros. Choose pairwise disjoint coordinate balls B_p for $p \in Z(X)$ such that X has no zero on ∂B_p. Perturb X inside the balls B_p, keeping it fixed near each ∂B_p, to obtain a vector field X' whose zeros are all nondegenerate. By the additivity of the local index,

$$\operatorname{ind}_p(X) = \sum_{q \in Z(X') \cap B_p} \operatorname{ind}_q(X').$$

Summing over all $p \in Z(X)$ gives

$$\sum_{p \in Z(X)} \operatorname{ind}_p(X) = \sum_{q \in Z(X')} \operatorname{ind}_q(X')$$

Since X' is transverse to the zero section, the first part of the proof gives

$$\sum_{q \in Z(X')} \operatorname{ind}_q(X') = \chi(M)$$

Therefore

$$\sum_{p \in Z(X)} \operatorname{ind}_p(X) = \chi(M)$$

which completes the proof of the Poincaré–Hopf theorem. □

Remark. The cobordism appearing in the Poincaré–Hopf theorem is not a cobordism of the ambient manifolds. In particular, the Euler characteristic is not an oriented cobordism invariant of manifolds. Rather, the cobordism is the cobordism of zero loci of homotopic sections. For vector fields, these sections are sections of TM, and their zero loci represent the Euler class of TM.

References

[1] John W. Milnor. *Topology from the Differentiable Viewpoint*. Based on notes by David W. Weaver. Charlottesville: University Press of Virginia, 1965.